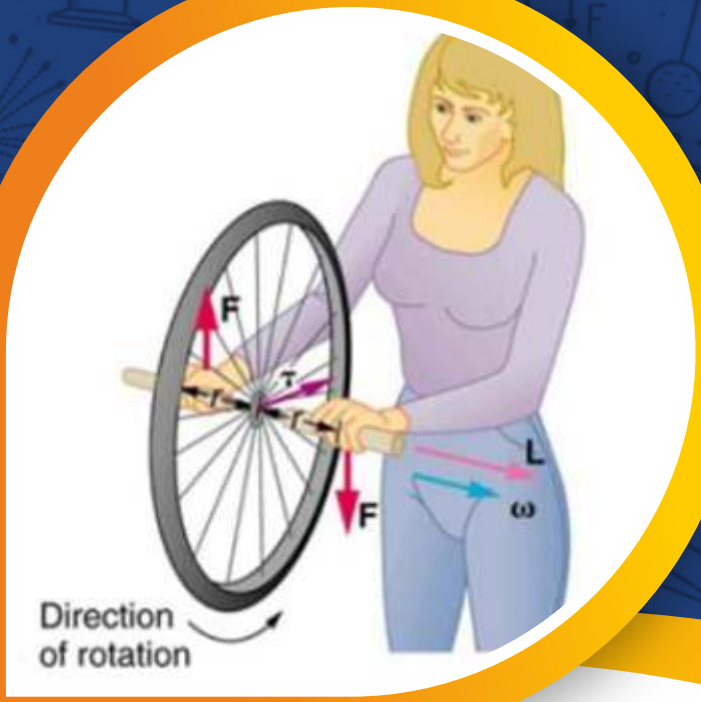
ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
path to success

PHYSICS

JEE (MAIN+ADVANCED)

NURTURE COURSE



Exercise

Rotational Dynamics

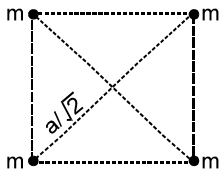
(English Medium)

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Moment of inertia

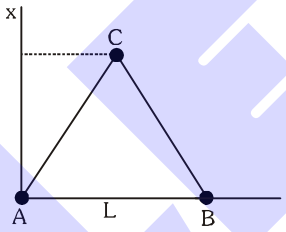
1. Four particles each of mass m are kept at the four corners of a square of edge a. Find the moment of inertia of the system about a line perpendicular to the plane of the square and passing through the centre of the square.



(A) $2ma^2$ (B) $4\ ma^2$ (C) $ma^2/2$ (D) $4\sqrt{2}ma^2$

RD0273

2. Three particles, each of mass m, are situated at the vertices of an equilateral triangle ABC of side L (figure). Find the moment of inertia of the system about the line AX perpendicular to AB in the plane of ABC.



(A) $\frac{5mL^2}{2}$ (B) $\frac{5mL^2}{4}$ (C) $\frac{3mL^2}{2}$ (D) $\frac{3mL^2}{4}$

RD0274

3. A diatomic molecule is formed by two atoms which may be treated as mass points m_1 and m_2 joined by a massless rod of length r. Then the moment of inertia of molecule about an axis passing through centre of mass and perpendicular to the rod is :

(A) zero (B) $(m_1 + m_2)r^2$ (C) $\left(\frac{m_1m_2}{m_1+m_2}\right)r^2$ (D) $\left(\frac{m_1+m_2}{m_1m_2}\right)r^2$

RD0275

4. Four point size dense bodies of same mass are attached at four corners of a light square frame. Identify the decreasing order of their moments of inertia about following axes.

(I) Passing through any side (II) Passing through opposite corners
(III) $\perp^r$ bisector of any side (IV) $\perp^r$ to the plane and passing through any corner
(A) III, IV, I, II (B) IV, III, I, II (C) III, II, IV, I (D) IV, III, II, I

RD0276

5. Three particles, each of mass 200 g, are kept at the corners of an equilateral triangle of side 10 cm. Find (i) the moment of inertia of the system about an axis joining two of the particles and (ii) moment of inertia of the system about an axis passing through one of the particles and perpendicular to the plane of the particles respectively :-

(A) 1.5 x 10^{-3} kg-m^2, 4.0 x 10^{-3} kg-m^2 (B) 3 x 10^{-3} kg-m^2, 2.0 x 10^{-3} kg-m^2
(C) 4.5 x 10^{-3} kg-m^2, 1.5 x 10^{-3} kg-m^2 (D) 2 x 10^{-3} kg-m^2, 1.5 x 10^{-3} kg-m^2

RD0277

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, VPE, COM & Coll & Rot. Dy\Eng\04_RD\01_Eng.p65

6. Particles of masses 1 g, 2 g, 3 g,....., 100 g are kept at the marks 1 cm, 2 cm, 3 cm,....., 100 cm respectively on a metre scale. Find the moment of inertia of the system of particles about a perpendicular bisector of the metre scale.

(A) 2.5 kg-m^2 (B) 1 kg-m^2 (C) 0.43 kg–m^2 (D) 2 kg-m^2

RD0278

7. Three bodies have equal mass m. Body A is solid cylinder of radius R, body B is a square lamina of side R, and body C is a solid sphere of radius R. Which body has the smallest moment of inertia about an axis passing through their centre of mass and perpendicular to the plane (in case of lamina)

(A) A (B) B (C) C (D) A and C both

RD0056

8. For the same total mass which of the following will have the largest moment of inertia about an axis passing through its centre of mass and perpendicular to the plane of the body

(A) a disc of radius a (B) a ring of radius a

(C) a square lamina of side 2a (D) four rods forming a square of side 2a

RD0057

9. A circular disc X of radius R is made from an iron plate of thickness t and another disc Y of radius 4R is made from an iron plate of thickness t/4. Then the relation between the moment of inertia I_X and I_Y is- **[AIEEE - 2003]**

(A) $I_Y = 32\ I_X$ (B) $I_Y = 16 I_X$ (C) $I_Y = I_X$ (D) $I_Y = 64\ I_X$

RD0061

10. A thin uniform rod of mass M and length L has its moment of inertia I_1 about its perpendicular bisector. The rod is bend in the form of a semicircular arc. Now its moment of inertia through the centre of the semi circular arc and perpendicular to its plane is I_2. The ratio of $I_1 : I_2$ will be _____

(A) < 1 (B) > 1 (C) = 1 (D) can't be said

RD0063

11. One solid sphere A and another hollow sphere B are of same mass and same outer radii. Their moment of inertia about their diameters are respectively I_A and I_B such that- **[AIEEE - 2004]**

(A) $I_A = I_B$ (B) $I_A > I_B$ (C) $I_A < I_B$ (D) $\frac{I_A}{I_B} = \frac{d_A}{d_B}$

where d_A and d_B are their densities. **RD0064**

12. A ring of mass 'm' and radius 'r' is melted and then moulded into a sphere . The moment of inertia of the sphere will be between about their symmetry axes)

(A) more than that of the ring (B) less than that of the ring

(C) equal to that of the ring (D) none of the above

RD0279

13. Find the moment of inertia of a solid cylinder of mass M and radius R about a line parallel to the axis of the cylinder and on the surface of the cylinder.

(A) $\frac{3}{2}MR^2$ (B) $\frac{5}{2}MR^2$ (C) $\frac{1}{2}MR^2$ (D) MR^2

RD0280

14. Moment of inertia of a hoop suspended from a peg about the peg is

(A) MR^2 (B) $\frac{MR^2}{2}$ (C) $2MR^2$ (D) $\frac{3MR^2}{2}$

RD0281

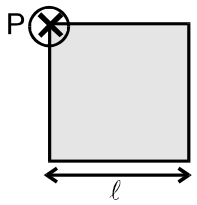
node06\B09C-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

15. Two identical spheres each of mass 1.20 kg and radius 10.0 cm are fixed at the ends of a light rod so that the separation between the centres is 50.0 cm. Find the moment of inertia of the system about an axis perpendicular to the rod passing through its middle point.

(A) 0.160 kg–m^2 (B) 0.320 kg–m^2 (C) 0.480 kg–m^2 (D) 0.640 kg–m^2

RD0282

16. Find the moment of inertia of a uniform square plate of mass M, edge of length 'ℓ' about its axis passing through P and perpendicular to it.



(A) $\frac{2}{3}M\ell^2$ (B) $\frac{5}{3}M\ell^2$ (C) $\frac{1}{6}M\ell^2$ (D) $\frac{1}{12}M\ell^2$

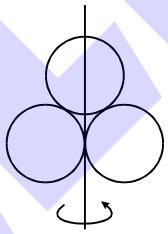
RD0283

17. Three identical uniform rods, each of length ℓ, are joined to form a rigid equilateral triangle. Find radius of gyration about an axis passing through a corner and perpendicular to the plane of the triangle.

(A) $\frac{\ell}{\sqrt{2}}$ (B) $\sqrt{2}\ell$ (C) $\frac{3\ell}{\sqrt{2}}$ (D) $\frac{\ell}{\sqrt{3}}$

RD0284

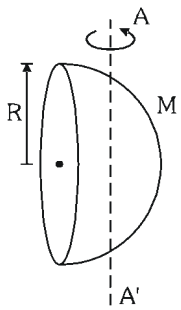
18. Three rings each of mass m and radius r are so placed that they touch each other. The radius of gyration of the system about the axis as shown in the figure is :



(A) $\sqrt{\frac{5}{3}}$ r (B) $\sqrt{\frac{5}{6}}$ r (C) $\sqrt{\frac{7}{3}}$ r (D) $\sqrt{\frac{7}{6}}$ r

RD0285

19. Find moment of inertia of a solid hemisphere of mass M shown in figure, about an axis AA' passing through its centre of mass.

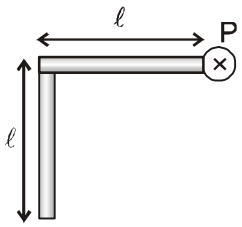


(A) $\frac{19}{320}MR^2$ (B) $\frac{21}{320}MR^2$ (C) $\frac{83}{320}MR^2$ (D) $\frac{1}{20}MR^2$

RD0286

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, VPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

20. Find the moment of inertia of the two uniform joint rods about an axis passing through P and perpendicular to plane of paper (mass of each rod = m and length of each rod = ℓ)

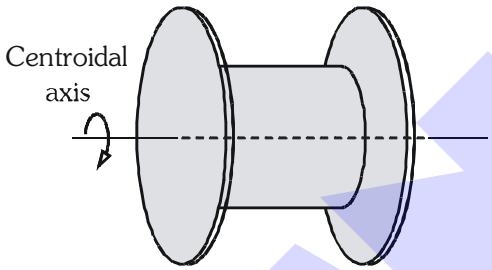


(A) $\frac{5m\ell^2}{3}$ (B) $\frac{2m\ell^2}{3}$

(C) $\frac{4m\ell^2}{3}$ (D) $2m\ell^2$

RD0287

21. Find moment of inertia about centroidal axis of a bobbin, which is constructed by joining coaxially two identical discs each of mass m and radius 2r to a cylinder of mass m and radius r as shown in the figure.



(A) $\frac{9}{2}mr^2$ (B) $\frac{5}{2}mr^2$ (C) $\frac{3}{2}mr^2$ (D) $\frac{7}{2}mr^2$

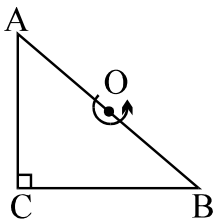
RD0288

22. Find moment of inertia of a uniform disc of mass m and radius r about one of its diameter.

(A) $\frac{1}{4}mr^2$ (B) $\frac{3}{4}mr^2$ (C) $\frac{5}{4}mr^2$ (D) $\frac{1}{2}mr^2$

RD0289

23. Find the moment of inertia of a plate cut in shape of a right angled triangle of mass M, about an axis perpendicular to the plane of the plate and passing through the mid point of side AB. (Side AC = BC = a)

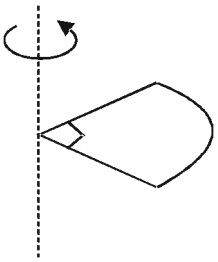


(A) $\frac{Ma^2}{12}$ (B) $\frac{Ma^2}{6}$

(C) $\frac{Ma^2}{3}$ (D) $\frac{2Ma^2}{3}$

RD0060

24. One quarter sector is cut from a uniform circular disc of radius R. This sector has mass M. It is made to rotate about a line perpendicular to its plane and passing through the center of the original disc. Its moment of inertia about the axis of rotation is **[IIT-JEE 2000]**

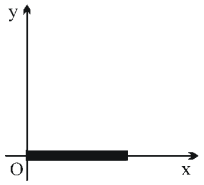


(A) $\frac{1}{2}$ MR^2 (B) $\frac{1}{4}$MR^2 (C) $\frac{1}{8}$ MR^2 (D) $\sqrt{2}$ MR^2

RD0059

node06\B09C-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

25. The figure shows a uniform rod lying along the x-axis. The locus of all the points lying on the xy-plane, about which the moment of inertia of the rod is same as that about O is :-



(A) an ellipse (B) a circle (C) a parabola (D) a straight line

RD0065

26. A rigid body can be hinged about any point on the x-axis. When it is hinged such that the hinge is at x, the moment of inertia is given by $I = 2x^2 - 12x + 27$. The x-coordinate of centre of mass is

(A) x = 2 (B) x = 0 (C) x = 1 (D) x = 3

RD0066

27. Find the moment of inertia of a pair of spheres, each having a mass m and radius r, kept in contact about the tangent passing through the point of contact.

(A) $\frac{14mr^2}{5}$ (B) $\frac{7mr^2}{5}$ (C) $\frac{4mr^2}{5}$ (D) $\frac{mr^2}{5}$

RD0290

28. Find the radius of gyration of a circular ring of radius r about a line perpendicular to the plane of the ring and passing through one of its particles.

(A) r (B) $\frac{r}{\sqrt{2}}$ (C) $\frac{r}{\sqrt{3}}$ (D) $\sqrt{2}\,r$

RD0291

29. Uniform square plate of mass 240 gram is made to rotate about an axis passing through any diagonal of plate. If its moment of inertia is 2×10^{-4} kgm^2 then its side length is

(A) 10 cm (B) 12 cm (C) 15 cm (D) 20 cm

RD0292

30. The mass of a thin circular plate is M and its radius is R. About an axis in the plane of plate at a perpendicular distance R/2 from centre of plate, its moment of inertia is

(A) $\frac{MR^2}{4}$ (B) $\frac{MR^2}{2}$ (C) $\frac{3MR^2}{4}$ (D) $\frac{3MR^2}{2}$

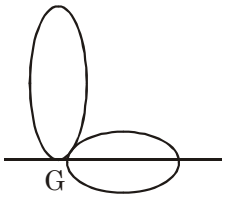
RD0293

31. Two identical circular plates each of mass M and radius R are attached to each other with their planes $\perp^r$ to each other one of the disc is attached to the centre of other disc. The moment of inertia of system about an axis passing through their centres and the point of contact is

(A) $\frac{MR^2}{4}$ (B) $\frac{5MR^2}{4}$ (C) $\frac{3}{4}MR^2$ (D) MR^2

RD0294

32. Two circular rings each of mass M and radius R are attached to each other at their rims and their planes perpendicular to each other as shown in the figure. The moment of inertia of the system about a diameter of one of the rings and passing through the point of contact is

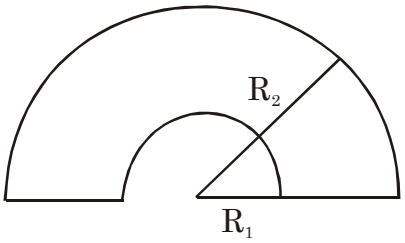


(A) $\frac{3}{2}MR^2$ (B) $\frac{3}{4}MR^2$ (C) $\frac{5}{2}MR^2$ (D) $\frac{5}{4}MR^2$

RD0295

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, VPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

33. Find moment of inertia of half disc of radius R_2 and mass M about its centre. A smaller half disc of radius R_1 is cut from this disc.

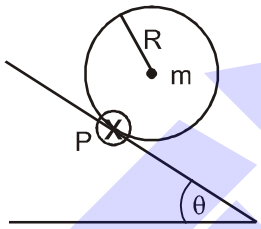


(A) $\frac{M}{4}\left(R_1^2+R_2^2\right)$ (B) $\frac{M}{8}\left(R_1^2+R_2^2\right)$ (C) $\frac{M}{16}\left(R_1^2+R_2^2\right)$ (D) $\frac{M}{32}\left(R_1^2+R_2^2\right)$

RD0296

Torque and equilibrium

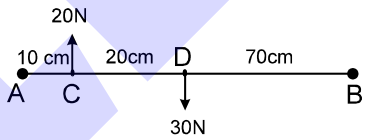
34. Find the torque of weight about the axis passing through point P.



(A) mgRsinθ (B) mgRcosθ (C) $\frac{mgR}{\cos\theta}$ (D) $\frac{mgR}{\sin\theta}$

RD0297

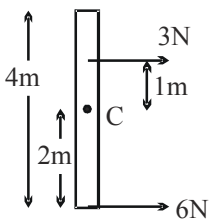
35. Find the distance of point of application of force from point A, when forces of 20 N & 30 N are acting on the rod as shown in figure.



(A) 70 cm (B) 50 cm (C) 10 cm (D) 60 cm

RD0298

36. A rod of mass 1 kg is kept on frictionless table and is acted upon by 2 forces as shown. We want to replace these 2 forces by a single force so that the effect on the rod is the same. How far from the centre C of the rod should we apply the force ?



(A) 0.5 m (B) 0.25 m (C) 1 m (D) 2 m

RD0299

node06\B09C-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Coll & Rot. Dy\Eng\04_RD\01_Eng.p65

37. A force $F = 2\hat{i} + 3\hat{j} - \hat{k}$ acts at a point (2, – 3, 1). Then magnitude of torque about point (0, 0, 2) will be:

(A) 6 unit (B) $3\sqrt{5}$ unit (C) $6\sqrt{5}$ unit (D) none of these

RD0300

38. If $\vec{F} = 3\hat{i}+5\hat{j}-2\hat{k}$ N acting at $7\hat{i} - 2\hat{j} + 5\hat{k}$, then find the torque about the point (0, – 1, 0).

(A) $-23\hat{i} + 29\hat{j} + 38\hat{k}$ (B) $-19\hat{i} + 29\hat{j} + 44\hat{k}$ (C) $-19\hat{i} - 29\hat{j} + 44\hat{k}$ (D) $-23\hat{i} - 29\hat{j} + 38\hat{k}$

RD0301

39. A uniform horizontal meter scale of mass m is suspended by two vertical strings attached to its two ends. A body of mass 2 m is placed on the 75 cm mark. Find the ratio of tensions in the strings.

(A) 3 : 2 (B) 1 : 2 (C) 2 : 1 (D) 1 : 3

RD0302

40. A weightless rod is acted on by upward parallel forces of 2N and 4N at ends A and B respectively. The total length of the rod is AB = 3m. To keep the rod in equilibrium a force of 6N should act in the following manner:

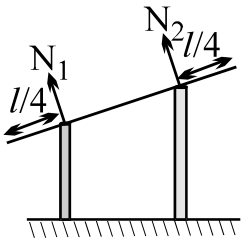
(A) Downwards at any point between A and B.

(B) Downwards at mid point of AB.

(C) Downwards at a point C such that AC = 1m.

(D) Downwards at a point D such that BD = 1m. **RD0067**

41. A uniform rod of length l is placed symmetrically on two walls as shown in figure. The rod is in equilibrium. If N_1 and N_2 are the normal forces exerted by the walls on the rod then :-



(A) $N_1 > N_2$

(B) $N_1 < N_2$

(C) $N_1 = N_2$

(D) N_1 and N_2 would be in the vertical directions **RD0069**

42. A uniform metre stick of mass 200 g is suspended from the ceiling through two vertical strings of equal lengths fixed at the ends. A small object of mass 20 g is placed on the stick at a distance of 70 cm from the left end. Find the tensions in the two strings.

(A) 1.04 N in the left string and 1.12 N in the right

(B) 1.04 N in the right string and 1.12 N in the left

(C) 2.08 N in the left string and 2.24 N in the right

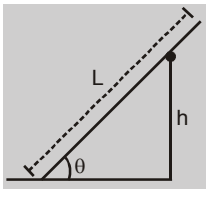
(D) 2.08 N in the right string and 2.24 N in the left

RD0303

43. A uniform rod of length L rests against a smooth roller as shown in figure. Find the friction coefficient between the ground and the lower end if the minimum angle that the rod can make with the horizontal is θ.

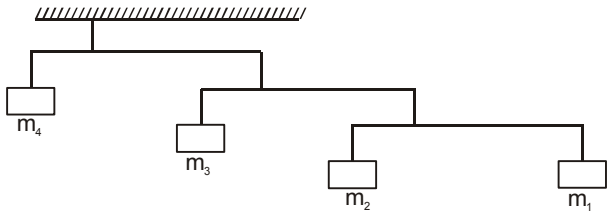
(A) $\dfrac{L\cos^2\theta\sin\theta}{2h - L\cos^2\theta\sin\theta}$ (B) $\dfrac{L\cos\theta\sin\theta}{2h - L\cos^2\theta\sin\theta}$

(C) $\dfrac{2L\cos\theta\sin^2\theta}{2h - L\cos^2\theta\sin\theta}$ (D) $\dfrac{L\cos\theta\sin^2\theta}{2h - L\cos^2\theta\sin\theta}$



RD0304

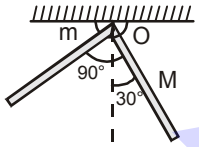
44. Figure shows an arrangement of masses hanging from a ceiling. In equilibrium, each rod is horizontal, has negligible mass and extends three times as far to the right of the wire supporting it as to the left. If mass m_4 is 48 kg then mass m_1 is equal to



(A) 1 kg (B) 2 kg (C) 3 kg (D) 4 kg

RD0305

45. Two uniform rods of equal length but different masses are rigidly joined to form an L–shaped body, which is then smoothly pivoted about O as shown. If in equilibrium the body is in the shown configuration, ratio M/m will be :



(A) 2 (B) 3 (C) $\sqrt{2}$ (D) $\sqrt{3}$

RD0306

46. The handle of a door is at a distance 40cm from axis of rotation. If a force 5N is applied on the handle in a direction 30° with plane of door, then the torque is

(A) 0.8 Nm (B) 1 Nm (C) 1.6 Nm (D) 2 Nm

RD0307

47. A uniform rod is 4m long and weight 10kg. If it is supported on a knife edge at one meter from the end, what weight placed at that end keeps the rod horizontal.

(A) 8kg (B) 10kg (C) 12kg (D) 15kg

RD0308

48. A metallic rod of mass 20 kg and of uniform thickness rests against a wall while the lower end of rod is in contact with rough floor. The rod makes an angle 60° with floor. If the weight of rod produces a torque 150 Nm about its lower end,the length of rod ($g = 10\ ms^{-2}$)

(A) 1.5 m (B) 2 m (C) 3 m (D) 4 m

RD0309

49. A roller of mass 300 kg and of radius 50 cm lying on horizontal floor is resting against a step of height 20 cm. The minimum horizontal force to be applied on the roller passing through its centre to turn the roller on to the step is

(A) 980 N (B)1960N (C) 2940 N (D) 3920 N

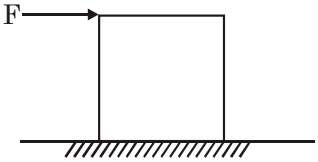
RD0310

50. A metallic cube of side length 1.5 m and of mass 3.2 ton is on horizontal rough floor. The minimum horizontal force that should be applied on the cube at a height 1.2 m from that floor to turn the cube about its lower edge is

(A) 1.96×10^3 N (B) 4.9×10^3 N (C) 1.96×10^4 N (D) 4.9×10^4N

RD0311

51. A cubical block of mass m and side L rests on a rough horizontal surface with coefficient of friction μ. A horizontal force F is applied on the block as shown. If the coefficient of friction is sufficiently high so that the block does not slide before toppling, the minimum force required to topple the block is:



(A) mg/4 (B) infinitesimal (C) mg/2 (D) mg (1 – u)

RD0312

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

52. A particle of mass m is projected with a speed u at an angle θ with the horizontal. Find the torque of the weight of the particle about the point of projection when the particle is at the highest point.
(A) $mu^2 \sin\theta \cos\theta$ perpendicular to the plane of motion
(B) $mu^2 \cos\theta$ perpendicular to the plane of motion
(C) $mu^2 \sin\theta \cos\theta$ perpendicular in the plane of motion
(D) $mu^2 \cos\theta$ perpendicular in the plane of motion

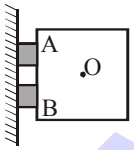
RD0313

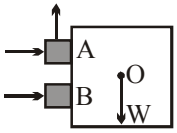
53. A cubical block of mass m and edge a slides down a rough inclined plane of inclination θ with a uniform speed. Find the torque of the normal force acting on the block about its centre.

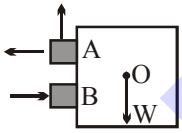
(A) Zero (B) $\frac{1}{2}mg\,a\cos\theta$ (C) $\frac{1}{2}mg\,a\sin\theta$ (D) $\frac{mga}{2}$

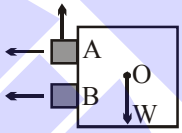
RD0314

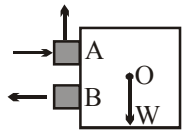
54. A vertical rectangular door with its centre of gravity at O (see figure) is fixed on two hinges A and B along one vertical length side of the door. The entire weight of the door is supported by the hinge A. Then the free body force diagram for the door (the arrows indicate the direction of the forces) is :-



(A) 

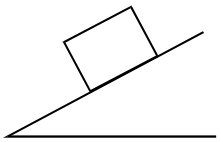
(B) 

(C) 

(D) 

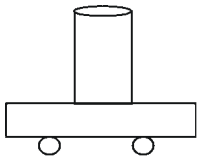
RD0070

55. A homogeneous cubical brick lies motionless on a rough inclined surface. The half of the brick which applies greater pressure on the plane is :
(A) left half
(B) right half
(C) both applies equal pressure
(D) the answer depend upon coefficient of friction



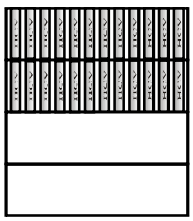
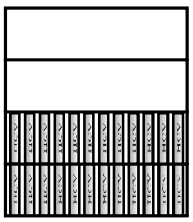
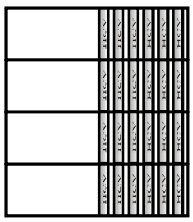
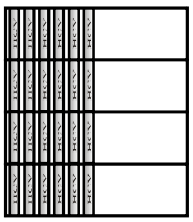
RD0074

56. A uniform 2 kg cylinder rests on a laboratory cart as shown. The coefficient of static friction between the cylinder and the cart is 0.5. If the cylinder is 4 cm in diameter and 10 cm in height, which of the following is closest to the maximum acceleration of the cart such that cylinder neither slips nor tips over?



(A) 2 m/s² (B) 4 m/s² (C) 5 m/s² (D) 6 m/s² **RD0075**

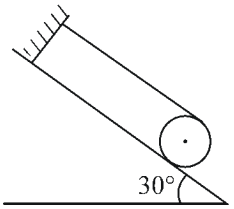
57. Same number of books are placed in four book cases as shown. Which bookcase is most likely to topple forward if pulled a little at the top towards right :

(A) 
(B) 
(C) 
(D) 

RD0073

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Collision & Rot. Dy\Eng\04_RD\01_Eng.p65

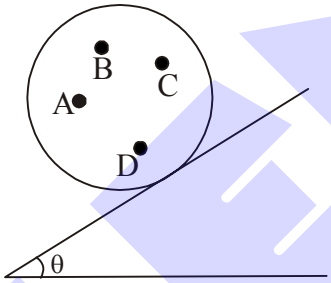
58. A thin hoop of weight 500 N and radius 1 m rests on a rough inclined plane as shown in the figure. The minimum coefficient of friction needed for this configuration to be in equilibrium is:



(A) $\frac{1}{3\sqrt{3}}$ (B) $\frac{1}{\sqrt{3}}$ (C) $\frac{1}{2}$ (D) $\frac{1}{2\sqrt{3}}$

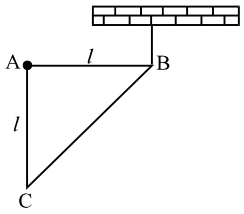
RD0071

59. A non uniform sphere can be kept on a rough inclined plane so that it is in equilibrium. In the figure below the dots represents location of centre of mass. In which one of the positions can sphere be in equilibrium?



RD0072

60. A right triangular plate ABC of mass m is free to rotate in the vertical plane about a fixed horizontal axis through A. It is supported by a string such that the side AB is horizontal. The reaction at the support A in equilibrium is:



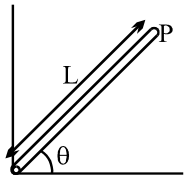
(A) $\frac{mg}{3}$ (B) $\frac{2mg}{3}$

(C) $\frac{mg}{2}$ (D) mg

RD0068

Fixed axis

61. A uniform flag pole of length L and mass M is pivoted on the ground with a frictionless hinge. The flag pole makes an angle θ with the horizontal. The moment of inertia of the flag pole about one end is $(1/3)ML^2$. If it starts falling from the position shown in the accompanying figure, the linear acceleration of the free end of the flag pole (labeled P) immediately after it starts falling off would be:

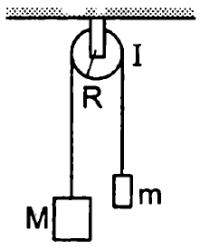


(A) (2/3) gcosθ (B) (2/3) g (C) g (D) (3/2) gcosθ

RD0077

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

62. The pulley shown in figure has a moment of inertia I about its axis and its radius is R. Find the magnitude of the acceleration of the two blocks. Assume that the string is light and does not slip on the pulley.

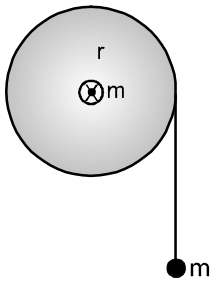


(A) $\frac{(M-m)gR^2}{I+(M+m)R^2}$ (B) $\frac{(M+m)gR^2}{I+(M-m)R^2}$

(C) $\frac{(M-m)gR^2}{I-(M+m)R^2}$ (D) $\frac{(M+m)gR^2}{I-(M+m)R^2}$

RD0315

63. A uniform disc of mass m, radius r and a point mass m are arranged as shown in the figure. The acceleration of point mass is: (Assume there is no slipping between pulley and thread and the disc can rotate smoothly about a fixed horizontal axis passing through its centre and perpendicular to its plane)



(A) $\frac{g}{2}$ (B) $\frac{g}{3}$

(C) $\frac{2g}{3}$ (D) none of these

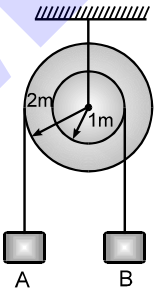
RD0316

64. A nearly massless rod is pivoted at one end so that it can swing freely as a pendulum. Two masses 2m and m are attached to it at distances b and 3b respectively from the pivot. The rod is held horizontal and then released. The angular acceleration of the rod at the instant it is released is

(A) bg (B) $\frac{b}{g}$ (C) gb + 5 (D) $\frac{5g}{11b}$

RD0317

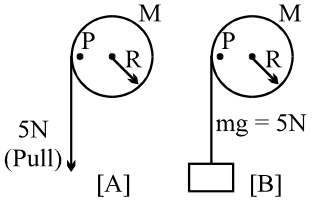
65. In the pulley system shown, if radii of the bigger and smaller pulley are 2 m and 1 m respectively and the acceleration of block A is 5 m/s^2 in the downward direction, then the acceleration of block B will be:



(A) 0 m/s^2 (B) 5 m/s^2 upwards (C) 10 m/s^2 upwards (D) 5/2 m/s^2 upwards

RD0318

66. A uniform disc of mass M = 2.50 kg and radius R = 0.20m is mounted on an axle supported on fixed frictionless bearings. A light cord wrapped around the rim is pulled with a force 5N. On the same system of pulley and string, instead of pulling it down, a body of weight 5N is suspended. If the first process is termed A and the second B, the tangential acceleration of point P will be



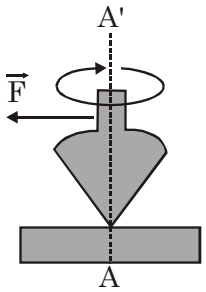
(A) equal in the processes A and B. (B) greater in process A than in B.

(C) greater in process B than in A. (D) independent of the two processes.

RD0319

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, VPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

67. The top in the figure has a moment of inertia equal to 4.0×10^{-4} kg m^2 and is initially at rest. It is free to rotate about the stationary axis AA^1. A string wrapped around a peg along the axis of the top is pulled in such a manner as to maintain a constant tension of 5.57N . If the string does not slip while it is unwound from the peg. what is the angular speed of the top after 80.0cm of string has been pulled off the peg.



(A) 30 rad/s (B) 142 rad/s (C) 149 rad/s (D) 120 rad/s

RD0320

68. A fly wheel of M.I. 6×10^{-2} kgm^2 is rotating with an angular velocity of 20 rads^{-1} . The torque required to bring it to rest in 4s is

(A) 1.6 N m (2) 0.6 Nm (3) 0.8Nm (4) 0.3 N m

RD0321

69. When 200 J of work is done on a fly wheel its frequency of rotation increases from 4Hz to 9Hz. The M.I. of the wheel about the axis of rotation is (nearly)

(A) 0.12 k g m^2 (B) 0.2 kg m^2 (C) 0.22 kg m^2 (D) 0.3 k gm^2

RD0322

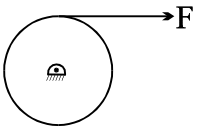
70. The moment of inertia of a wheel of radius 20 cm is 40 kgm^2 if a tangential force of 80N applied on the wheel, its rotational K.E. after 4s is

(A) 16.2 J (B) 51.2 J (C) 25.6 J (D) 24.8 J

RD0323

71. A pulley is hinged at the centre and a massless thread is wrapped around it. The thread is pulled with a constant force F starting from rest. As time increases,

(A) its angular velocity increases, but force on hinge remains constant
(B) its angular velocity remains same, but force on hinge increases
(C) its angular velocity increases and force on hinge increases
(D) its angular velocity remains same and force on hinge is constant



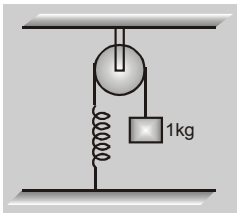
RD0078

72. A wheel of mass 10 kg and radius 20 cm is rotating at an angular speed of 100 rev/min when the motor is turned off. Neglecting the friction at the axle, calculate the force that must be applied tangentially to the wheel to bring it to rest in 10 revolutions.

(A) 1.74 N (B) 2.61 N (C) 0.44 N (D) 0.87 N

RD0324

73. The pulley shown in figure has a radius of 20 cm and moment of inertia 0.2 kg-m^2. The string going over it is attached at one end to a vertical spring of spring constant 50 N/m fixed from below, and supports a 1 kg mass at the other end. The system is released from rest with the spring at its natural length. Find the speed of the block when it has descended through 10 cm. Take g = 10 m/s^2.



(A) 0.1 m/s (B) 0.5 m/s (C) 0.25 m/s (D) 1 m/s

RD0325

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

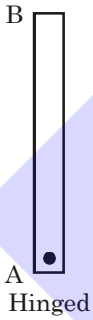
74. A thin rod of mass m and length l is suspended vertically as shown in the fig. If rod is initially at rest, then find minimum angular velocity ω given to the rod so that it can complete rotation about the horizontal axis passing through upper end



(A) $\sqrt{\frac{5g}{\ell}}$ (B) $\sqrt{\frac{6g}{\ell}}$ (C) $\sqrt{\frac{10g}{\ell}}$ (D) $\sqrt{\frac{12g}{\ell}}$

RD0326

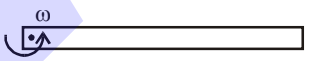
75. A uniform rod of mass m length ℓ is hinged at A. It is released from the given position so that it rotates about A in vertical plane. Which of the following represents the hinge force when rod becomes horizontal. (Rod is rotating clockwise) :-



(A) ← (B) ↙ (C) ↑ (D) ↖

RD0327

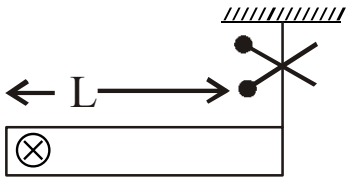
76. A rod of mass m and length ℓ lying on a smooth horizontal table is rotating in horizontal plane about a vertical axis passing through one end of rod as shown in the figure. Force applied by hinge on the rod is :-



(A) $m\omega^2\ell$ (B) $\frac{m\ell^2\omega}{2}$ (C) $\frac{m\omega^2\ell}{2}$ (D) $\frac{m\omega^2\ell}{4}$

RD0328

77. Calculate the hinge force when the rod becomes vertical. Initial situation is shown in figure and string is then cut. (hinge is frictionless)



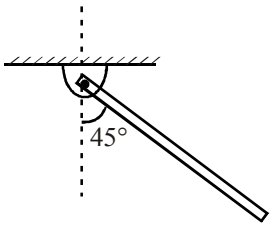
(A) $F_x = 0;\ F_y = \frac{3}{2}Mg$ (B) $F_x = Mg\ ;\ F_y = \frac{5}{2}Mg$

(C) $F_x = \frac{3}{2}Mg\ ;\ F_y = \frac{3}{2}Mg$ (D) $F_x = 0\ ;\ F_y = \frac{5}{2}Mg$

RD0329

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, VPE, COM & Coll & Rot. Dy\Eng\04_RD\01_Eng.p65

78. A uniform rigid rod hinged at one end is released from rest in the position shown in the vertical plane. Find the magnitude of reaction force (in N) at hinge just after its release. (Use : $M = 0.8kg.$, $\theta = 45°$, $g = 10\ m/s^2$)



(A) $\sqrt{34}$ N (B) 5N (C) $\sqrt{37}$ N (D) 6 N

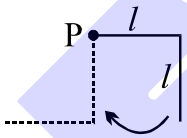
RD0330

79. A thin uniform rod XY of length 2l is hinged at one end X to the floor and stands vertical. When allowed to fall, the angular speed with which the rod strikes the floor is

(A) $\sqrt{\frac{3g}{4l}}$ (B) $\sqrt{\frac{3g}{2l}}$ (C) $\sqrt{\frac{3g}{l}}$ (D) $\sqrt{\frac{g}{2l}}$

RD0331

80. An L shaped thin uniform rod of total length $2l$ is free to rotate in a vertical plane about a horizontal axis at P as shown in the figure.The bar is released from rest. Neglect air and contact friction. The angular velocity at the instant it has rotated through 90° and reached the dotted position shown is



(A) zero (B) $\sqrt{\frac{6g}{5l}}$ (C) $\sqrt{\frac{3g}{5l}}$ (D) none

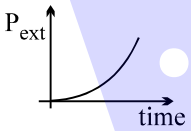
RD0332

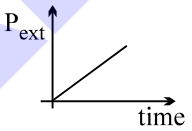
81. A constant power is supplied to a rotating disc. Angular velocity (ω) of disc varies with number of rotations (n) made by the disc as

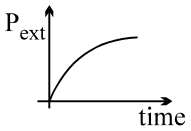
(A) $\omega \propto n^{1/3}$ (B) $\omega \propto n^{3/2}$ (C) $\omega \propto n^{2/3}$ (D) $\omega \propto n^2$

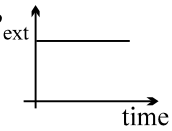
RD0333

82. A rod is hinged at its centre and rotated by applying a constant torque starting from rest. The power developed by the external torque as a function of time is :

(A) 

(B) 

(C) 

(D) 

RD0118

83. Electric power is used to speed up a centrifuge whose rotational inertia is 1.2 kg•m^2. Average 1.6 kW of power were used to make the centrifuge accelerate at a steady rate from rest to 18,000 rev/min. It the electricity use was 100 percent efficient, how much time (in min) is required to speed up the centrifuge? (Take $\pi^2 = 10$)

(A) 22.5 (B) 11.25 (C) 45 (D) 5.625 **RD0334**

Angular momentum & its conservation

84. A particle of mass 2 kg located at the position $(\hat{i} + \hat{j})$ m has a velocity $2(+\hat{i} - \hat{j} + \hat{k})$ m/s. Its angular momentum about z-axis in kg-m^2/s is:

(A) zero (B) +8 (C) 12 (D) –8

RD0079

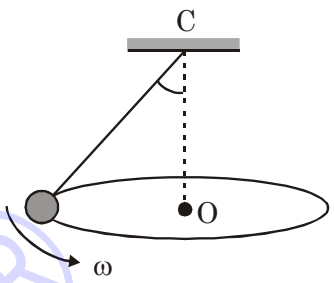
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

85. A particle of mass m is projected with speed u at an angle θ with the horizontal. Find angular momentum of particle about point of projection when particle again at same horizontal level :

(A) $\frac{mu^3 \sin\theta \cdot \sin 2\theta}{g}$ (B) $\frac{mu^3 \cos\theta \cdot \sin 2\theta}{g}$ (C) $\frac{mu^3 \sin\theta \cdot \sin 2\theta}{2g}$ (D) $\frac{mu^3 \sin\theta \cdot \cos 2\theta}{g}$

RD0335

86. A ball is being revolved in a circle by the means of a string as in the case of conical pendulum. About which of the points will the angular momentum of the ball be conserved. Assume that there is no air resistance.



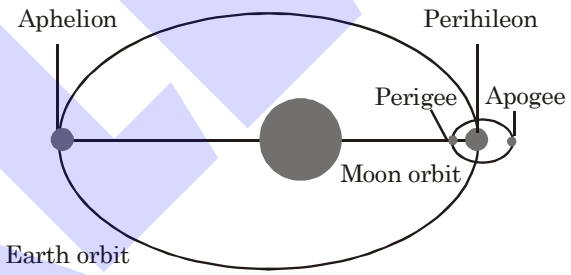
(A) C
(B) O
(C) Any point on line OC
(D) The angular momentum is not conserved about any point.

RD0336

87. A small disc sliding on a smooth horizontal surface strikes a smooth wall. The collision is partially elastic. About which of the points will the angular momentum of disc be conserved.
(A) only the Point of collision
(B) Any point on the normal to the wall drawn from the point of collision.
(C) The angular momentum is not conserved about any point.
(D) The angular momentum is conserved about all the points in space

RD0337

88. It is fairly common knowledge that earth revolves around the sun in an elliptical orbit. About which point will the angular momentum of the earth around the sun be conserved ?



(A) Aphelion (B) Perihileon
(C) Centre of the ellipse (D) Centre of the Sun

RD0338

89. A constant torque acting on a uniform circular wheel changes its angular momentum from A_0 to $4A_0$ in 4 seconds. The magnitude of this torque is-

(A) $\frac{3A_0}{4}$ (B) A_0 (C) $\frac{4A_0}{3}$ (D) $12A_0$

RD0339

90. A ring (m, r) is rotating about it's axis with angular momentum L. It's rotational kinetic energy will be :

(A) $\frac{L}{2mr^2}$ (B) $\frac{L^2}{2mr^2}$ (C) $\frac{L^2}{mr^2}$ (D) $\sqrt{2mL^2r^2}$

RD0340

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, VPE, COM & Coll & Rot. Dy\Eng\04_RD\01_Eng.p65

91. A spinning ice skater can increase his rate of rotation by bringing his arms and free leg closer to his body. How does this procedure effect the skater's angular momentum and kinetic energy and what is the work done by the skater?
(A) angular momentum remains the same while kinetic energy increases and work done is positive.
(B) angular momentum remains the same while kinetic energy decreases and work done is negative.
(C) both angular momentum and kinetic energy remain the same and work done is zero.
(D) angular momentum increases while kinetic energy remains the same and work done may be positive or negative.
RD0082

92. If earth shrinks then the duration of day
(A) increases (B) decreases
(C) remains same (D) first increases then decreases to initial value
RD0341

93. A circular disc is rotating in horizontal plane about vertical axis passing through its centre without friction with a person standing on the disc at its edge. If the person gently walks to centre of disc then its angular velocity
(A) increases (B) decreases (C) does not change (D) becomes zero
RD0342

94. A circular ring of mass M is rotating about its own axis in horizontal plane at an angular velocity ω. If two point size bodies each of mass m, are gently attached to the rim of ring at two ends of its diameter, then the angular velocity of ring is

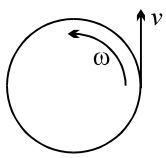
(A) $\frac{M\omega}{M+2m}$ (B) $\frac{2m\omega}{M+2m}$ (C) $\frac{m\omega}{M+2m}$ (D) $\frac{2M\omega}{M+2m}$
RD0343

95. A circular disc of mass 4 kg and of radius 10 cm is rotating about its natural axis at the rate of 5 rad/sec. its angular momentum is
(A) 0.25 kgm^2s^{-1} (B) 0.1 $kgm^2 s^{-1}$ (C) 2.5 kgm^2s^{-1} (D) 0.2 kgm^2s^{-1}
RD0344

96. $\vec{F} = a\hat{i} + 3\hat{j} + 6\hat{k}$ and $\vec{r} = 2\hat{i} - 6\hat{j} - 12\hat{k}$. The value of 'a' for which the angular momentum is conserved is
(A) –1 (B) 0 (C) 1 (D) 2
RD0345

97. A child with mass *m* is standing at the edge of a disc with moment of inertia I, radius R, and initial angular velocity ω. See figure given below. The child jumps off the edge of the disc with tangential velocity *v* with respect to the ground. The new angular velocity of the disc is

(A) $\sqrt{\frac{I\omega^2 - mv^2}{I}}$ (B) $\sqrt{\frac{(I+mR^2)\omega^2 - mv^2}{I}}$



(C) $\frac{I\omega - mvR}{I}$ (D) $\frac{(I+mR^2)\omega - mvR}{I}$
RD0083

98. A wheel of moment of inertia 0.500 kg-m^2 and radius 20.0 cm is rotating about its axis at an angular speed of 20 rad/s. It picks up a stationary particle of mass 200 g at its edge. Find the new angular speed of the wheel.
(A) 19.7 rad/s (B) 9.85 rad/s (C) 59.1 rad/s (D) 6.57 rad/s
RD0346

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

99. A driver having a moment of inertia of 6.0 kg-m^2 about an axis through its centre of mass rotates at an angular speed of 2 rad/s about this axis. If he folds his hands and feet to decrease the moment of inertia to 5.0 kg-m^2, what will be the new angular speed ?
(A) 1.2 rad/s (B) 2.4 rad/s (C) 4.8 rad/s (D) 0.8 rad/s

RD0347

100. A boy is seated in a revolving chair revolving at an angular speed of 120 revolutions per minute. Two heavy balls form part of the revolving system and the boy can pull the balls closer to himself or may push them apart. If by pulling the balls closer, the boy decreases the moment of inertia of the system from 6 kg-m^2 to 2 kg-m^2, what will be the new angular speed ?
(A) 240 rev/minute (B) 120 rev/minute (C) 180 rev/minute (D) 360 rev/minute

RD0348

101. A wheel of moment of inertia 0.10 kg-m^2 is rotating about a shaft at an angular speed of 160 rev/minute. A second wheel is set into rotation at 300 rev/minute and is coupled to the same shaft so that both the wheels finally rotate with a common angular speed of 200 rev/minute. Find the moment of inertia of the second wheel.
(A) 0.04 kg-m^2 (B) 0.08 kg-m^2 (C) 0.12 kg-m^2 (D) 0.16 kg-m^2

RD0349

102. A kid of mass M stands at the edge of a platform of radius R which can be freely rotated about its axis. The moment of inertia of the platform is I. The system is at rest when a friend throws a ball of mass m and the kid catches it. If the velocity of the ball is v horizontally along the tangent to the edge of the platform when it was caught by the kid, find the angular speed of the platform after the event.

(A) $\frac{mvR}{I+(M+m)R^2}$ (B) $\frac{mvR}{I+(M-m)R^2}$ (C) $\frac{mvR}{I-(M-m)R^2}$ (D) $\frac{2mvR}{I+(M+m)R^2}$

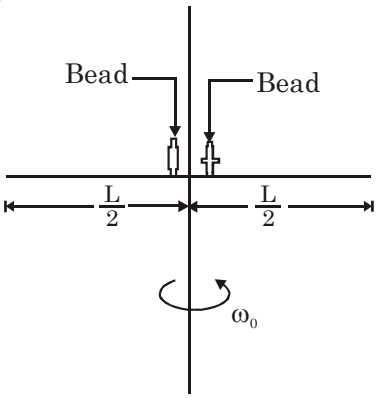
RD0350

103. A circular disc is rotating without friction about its natural axis with an angular velocity ω. Another circular disc of same material and thickness but half the radius is gently placed over it coaxially. The angular velocity of composite disc will be

(A) $\frac{4\omega}{3}$ (B) $\frac{8\omega}{9}$ (C) $\frac{7\omega}{8}$ (D) $\frac{16\omega}{17}$

RD0351

104. A smooth uniform rod of length L and mass M has two identical beads of negligible size, each of mass m, which can slide freely along the rod. Initially the two beads are at the centre of the rod and the system is rotating with angular velocity ω_0 about its axis perpendicular to the rod and passing through its mid point (see figure). There are no external forces. When the beads reach the ends of the rod, the angular velocity of the system is **[JEE - 1988]**

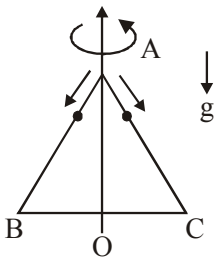


(A) $\frac{M\omega_0}{M+3m}$ (B) $\frac{M\omega_0}{M+6m}$ (C) $\frac{(M+6m)\omega_0}{M}$ (D) ω_0

RD0352

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, VPE, COM & Collision & Rot. Dy\Eng\04_RD\01_Eng.p65

105. An equilateral triangle ABC formed from a uniform wire has two small identical beads initially located at A. The triangle is set rotating about the vertical axis AO. Then the beads are released from rest simultaneously and allowed to slide down, one along AB and the other along AC as shown. Neglecting frictional effects, the quantities that are conserved as the beads slide down, are



(A) Angular velocity and total energy (kinetic and potential)
(B) Total angular momentum and total energy
(C) Angular velocity and moment of inertia about the axis of rotation.
(D) Total angular momentum and moment of inertia about the axis of rotation.

[IIT-JEE 2000]

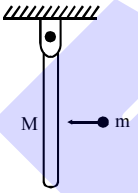
RD0084

106. A mass m is moving at speed v perpendicular to a rod of length d and mass $M = 6m$ which pivots around a frictionless axle running through its centre. It strikes and sticks to the end of the rod. The moment of inertia of the rod about its centre is $Md^2/12$. Then the angular speed of the system right after the collision is

(A) $2v/d$ (B) $2v/(3d)$ (C) v/d (D) $3v/(2d)$

RD0101

107. A uniform rod of mass M is hinged at its upper end. A particle of mass m moving horizontally strikes the rod at its mid point elastically. If the particle comes to rest after collision find the value of M/m =?



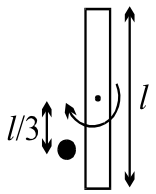
(A) 3/4 (B) 4/3 (C) 2/3 (D) 3/2

RD0353

108. A uniform rod of length l and mass M rotating about a fixed vertical axis on a smooth horizontal table. It elastically strikes a particle placed at a distance $l/3$ from its axis and stops. Mass of the particle is

(A) 3M (B) $\frac{3M}{4}$

(C) $\frac{3M}{2}$ (D) $\frac{4M}{3}$



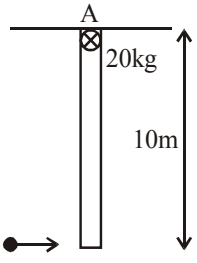
RD0100

109. A uniform thin rod of mass 'm' and length L is hinged about its upper end, and is free to swing in a vertical plane. A tiny ball of mass m/4 hits the lower end of the rod. The rod was initially vertical and the ball was moving horizontally with velocity v_0. If the ball stops just after collision, then 'e'=

(A) 0.75 (B) 0.25 (C) 0.5 (D) none

RD0354

110. A rod of mass 20 kg & length 10m is hinged at A & hanging vertically. A bullet of mass 5kg moving with velocity 10 m/s sticks to one end of rod. Find angular velocity of rod just after the colliding particle sticks to it :-



(A) $\frac{7}{3}$ rad / sec (B) $\frac{3}{7}$ rad / sec

(C) $\frac{3}{10}$ rad / sec (D) 3 rad/sec

RD0355

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

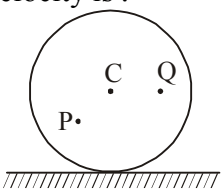
Rolling Kinematics

111. A disc of radius 10 cm is moving with its centre's velocity = 1 m/s rightwards, at a certain moment, on a flat horizontal surface. At this moment, its angular velocity is 20 rad/s, anticlockwise. Then, the distance of a point on the disc (from its centre) whose instantaneous velocity is zero is

(A) 5 cm (B) 10/3 cm (C) 5/3 cm (D) 2.5 cm

RD0356

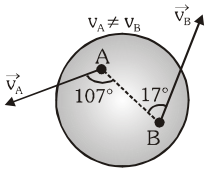
112. A disc is rolling without slipping with angular velocity ω. P and Q are two points equidistant from the centre C. The order of magnitude of velocity is :- **[IIT-JEE 2004]**

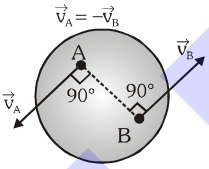


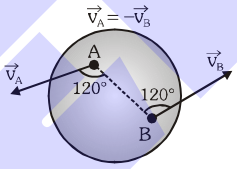
(A) $v_Q > v_C > v_P$ (B) $v_P > v_C > v_Q$ (C) $v_P = v_C$, $v_Q = v_C/2$ (D) $v_P < v_C > v_Q$

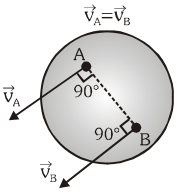
RD0086

113. A laminar rigid body is confined to move in its own plane. At some instant velocities of any two points on the body are shown in following figures. Which one of the following physical situation cannot be possible?

(A) 

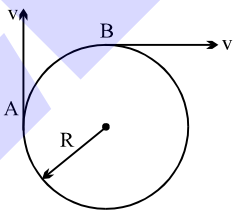
(B) 

(C) 

(D) 

RD0357

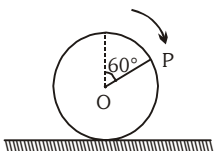
114. Two points of a rigid body are moving as shown. The angular velocity of the body is:



(A) $\frac{\upsilon}{2R}$ (B) $\frac{\upsilon}{R}$ (C) $\frac{2\upsilon}{R}$ (D) $\frac{2\upsilon}{3R}$

RD0055

115. A wheel of radius R = 0.1 m is rolling without slipping on a horizontal surface as shown in the figure. Centre of the wheel moves with a constant speed √3 m/s. The speed of the point P with respect to ground is

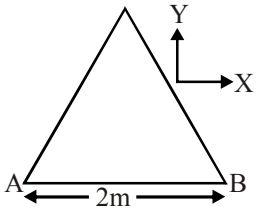


(A) 2√3 m/s (B) zero (C) 3 m/s (D) √3 m/s

RD0358

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, VPE, COM & Collision & Rot. Dy\Eng\04_RD\01_Eng.p65

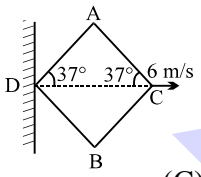
116. A rigid equilateral triangular plate ABC of side 2 m, is in motion in the x-y plane. At the instant shown in the figure, the point B has velocity $\vec{v}_B = (3\hat{i} + 8\hat{j})\,m/s$ and the plate has angular velocity $\vec{\omega} = 2\hat{k}\ r/s$. Find the speed of point A.



(A) 5 m/s (B) 4 m/s (C) 3 m/s (D) None of these

RD0359

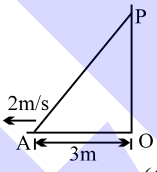
117. A rhombus ABCD is shown in the figure. The sides of the rhombus can rotate about vertex A, B, C & D. The vertex C is moving with a velocity of 6 m/s in horizontal direction. Determine the velocity of vertex A.



(A) 4.8 m/s (B) 4.5 m/s (C) 5 m/s (D) none of these

RD0360

118. A ladder AP of length 5 m is inclined to a vertical wall is slipping over a horizontal surface with velocity of 2 m/s, when A is at a distance 3m from ground what is the velocity of C.M. at this moment



(A) 1.25 m/s (B) 0 m/s (C) 1 m/s (D) 2 m/s

RD0361

119. A disc is rotating with angular velocity (ω) about its axis (without any translation push) on a smooth surface :

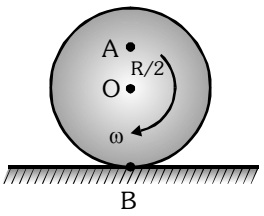
The directions and magnitudes of velocity at points B and A are :-

(A) $V_A = +\frac{\omega R}{2}$ (Towards right), $V_B = -\omega R$ (Towards left)

(B) $V_A = -\omega R$ (Towards right), $V_B = \frac{\omega R}{2}$ (Towards left)

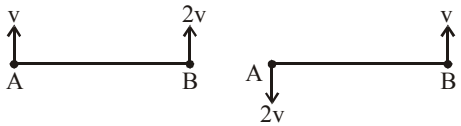
(C) $V_A = +\frac{\omega R}{4}$ (Towards right), $V_B = -\omega R$ (Towards left)

(D) $V_A = +\frac{\omega R}{2}$ (Towards right), $V_B = \frac{\omega R}{2}$ (Towards left)



RD0362

120. A rod AB is a uniform rod. The end points on the rod are given velocities is two cases as shown. Find the ratio of distance of instantaneous axis of rotation from A in both the cases :-

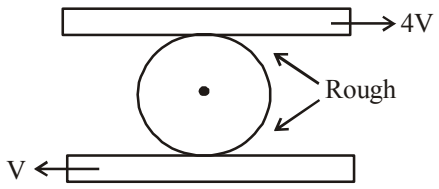


(A) 2 : 3 (B) 2 : 1 (C) 1 : 1 (D) 3 : 2

RD0363

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

121. The arrangement shows a solid cylinder of radius R placed between two planks. There is no slipping at rough surfaces. The planks are given a horizontal constant velocity as shown. The angular velocity of the cylinder is:



(A) $\frac{5V}{4R}$ (B) $\frac{2V}{3R}$ (C) $\frac{3V}{4R}$ (D) $\frac{5V}{2R}$

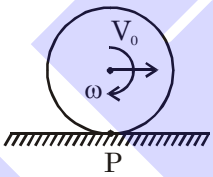
RD0364

122. A body kept on a smooth horizontal surface is pulled by a constant horizontal force applied at the top point of the body. If the body rolls purely on the surface, its shape can be :

(A) thin pipe
(B) uniform cylinder
(C) uniform sphere
(D) thin spherical shell

RD0092

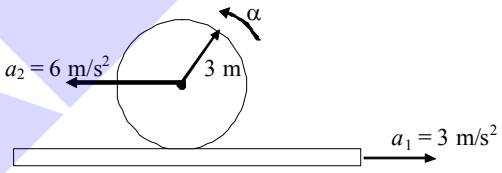
123. A disc is rolling uniformly on ground surface as shown in the figure. Velocity of centre is V_0 and angular velocity is ω. Then acceleration of point P on the disc is:



(A) zero
(B) $r\omega^2$ vertically upward
(C) $r\omega^2$ vertically downward
(D) None

RD0365

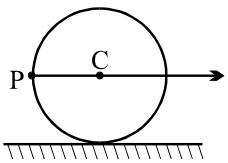
124. In the following figure, a sphere of radius 3 m rolls on a plank. The accelerations of the sphere and the plank are indicated. The value of α is



(A) 3 rad/s^2
(B) 6 rad/s^2
(C) 3 rad/s^2 (opposite to the direction shown in figure)
(D) 1 rad/s^2

RD0087

125. A disc of radius R is rolling purely on a flat horizontal surface, with a constant angular velocity. The angle between the velocity and acceleration vectors of point P is



(A) zero (B) 45° (C) 135° (D) $\tan^{-1}(1/2)$

RD0088

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Collision & Rot. Dy\Eng\04_RD\01_Eng.p65

Rolling dynamics

126. A cylinder rolls up an inclined plane, reaches some height, and then rolls down (without slipping throughout these motions). The directions of the frictional force acting on the cylinder are
(A) Up the incline while ascending and down the incline while descending **[IIT-JEE 2002]**
(B) Up the incline while ascending as well as descending
(C) down the incline while ascending and up the incline while descending
(D) down the incline while ascending as well as descending **RD0090**

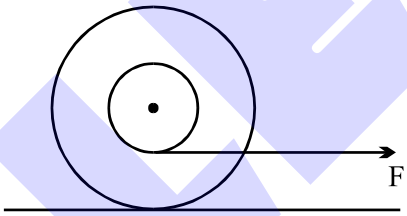
127. A sphere of mass m rolls without slipping on an inclined plane of inclination θ. What should be the minimum coefficient of static friction to support pure rolling ?

(A) $\mu > \frac{2}{7}\tan\theta$ (B) $\frac{2}{5}\tan\theta$ (C) $\frac{2}{3}\tan\theta$ (D) $\tan\theta$

RD0366

128. Two solid cylinders P and Q of same mass and same radius start rolling down a fixed inclined plane from the same height at the same time. Cylinder P has most of its mass concentrated near its surface, while Q has most of its mass concentrated near the axis. Which statement(s) is(are) correct?
(A) Both cylinders P and Q reach the ground at the same time. **[IIT-JEE 2012]**
(B) Cylinder P has larger acceleration than cylinder Q.
(C) Both cylinders reach the ground with same translational kinetic energy.
(D) Cylinder Q reaches the ground with larger angular speed. **RD0179**

129. Inner and outer radii of a spool are r and R respectively. A thread is wound over its inner surface and placed over a rough horizontal surface. Thread is pulled by a force F as shown in fig. then in case of pure rolling



(A) Thread unwinds, spool rotates anticlockwise and friction act leftwards
(B) Thread winds, spool rotates clockwise and friction acts leftwards
(C) Thread winds, spool moves to the right and friction act rightwards
(D) Thread winds, spool moves to the right and friction does not come into existence.

RD0091

130. A solid sphere, a hollow sphere and a ring are released from top of an inclined plane (frictionless) so that they slide down the plane. Then maximum acceleration down the plane is (no rolling):
(A) solid sphere (B) hollow sphere (C) ring (D) same for all

RD0367

131. In the case of following rolling body translatory and rotational kinetic energies are equal for
(A) circular ring (B) circular plate (C) solid sphere (D) solid cylinder

RD0368

132. A solid sphere rolls down without slipping from rest on a 30° incline. Its linear acceleration is
(A) 5g/7 (B) 5g/14 (C) 2g/3 (D) g/3

RD0369

133. A circular ring starts rolling down on an inclined plane from its top. Let v be velocity of its centre of mass on reaching the bottom of inclined plane. If a block starts sliding down on an identical inclined plane but smooth, from its top, then the velocity of block on reaching the bottom of inclined plane is

(A) $\frac{v}{2}$ (B) 2v (C) $\frac{v}{\sqrt{2}}$ (D) $\sqrt{2}v$

RD0370

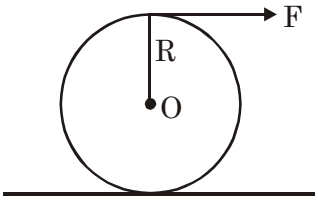
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

134. A small sphere of radius R rolls without slipping inside a large hemispherical bowl of radius R . The sphere starts from rest at the top point of the hemisphere. What fraction of the total energy is rotational when the small sphere is at the bottom of the hemisphere

(A) $\frac{7}{5}$ (B) $\frac{2}{7}$ (C) $\frac{5}{7}$ (D) $\frac{7}{10}$

RD0371

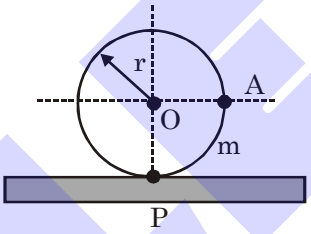
135. A tangential force F acts at the top of a thin spherical shell of mass m and radius R. Find the acceleration of the shell if it rolls without slipping.



(A) $\frac{6F}{5m}$ (B) $\frac{6m}{5F}$ (C) $\frac{3m}{5F}$ (D) $\frac{5m}{6F}$

RD0372

136. A particle of mass 'm' is rigidly attached at 'A' to a ring of mass '3m' and radius 'r'. The system is released from rest and rolls without sliding. The angular acceleration of ring just after release is



(A) $\frac{g}{4r}$ (B) $\frac{g}{6r}$ (C) $\frac{g}{8r}$ (D) $\frac{g}{2r}$

RD0373

137. A hoop and a solid cylinder have the same mass and radius. They both roll, without slipping, on a horizontal surface. If their kinetic energies are equal

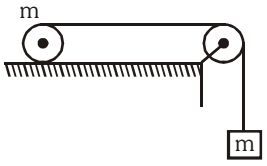
(A) the hoop has a greater translational speed than the cylinder

(B) the cylinder has a greater translational speed than the hoop

(C) the hoop and the cylinder have the same translational speed

(D) the hoop has a greater rotational speed than the cylinder **RD0094**

138. In the given figure a ring of mass m is kept on a horizontal surface while a body of equal mass 'm' attached through a string, which is wounded on the ring. When the system is released the ring rolls without slipping. Consider the following statements and choose the correct option.



(i) acceleration of the centre of mass of ring is $\frac{g}{3}$.

(ii) acceleration of the hanging particle is $\frac{2g}{3}$.

(iii) frictional force (on the ring) acts along forward direction.

(iv) frictional force (on the ring) acts along backward direction.

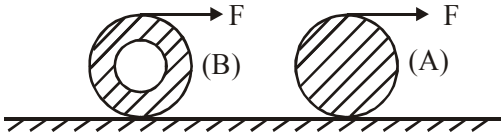
(A) statement (i) and (ii) only. (B) statement (ii) and (iii) only.

(C) statement (iii) and (iv) only. (D) none of these.

RD0374

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

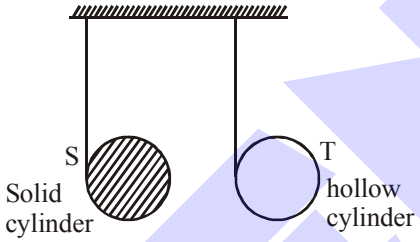
139. Disc A and disc B with a hole have equal mass and radius and a string is wrapped around them as shown. Equal force F is applied on the string of each body. Friction is sufficient for rolling. After time t velocity of A is v_A and that of B is v_B and kinetic energy k_A & k_B, then :-



(A) $v_A = v_B$ (B) $v_A < v_B$ (C) $k_A > k_B$ (D) $k_A < k_B$

RD0375

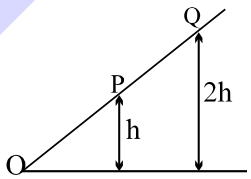
140. A solid cylinder S and a hollow cylinder T of same mass and same outer radius have a thin thread wound over them. One end of each thread is fixed to the same ceiling.They are initially held at rest with their axis horizontal and parallel to each other, at same height above ground. If the cylinders are released simultaneously with strings taut then the correct statement is :



(A) S will reach ground earlier than T
(B) T will reach ground earlier than S
(C) Both will reach ground together
(D) For small height S will reach earlier and for large height T will reach earlier.

RD0376

141. A ball rolls down an inclined plane as shown in figure. The ball is first released from rest from P and then later from Q. Which of the following statement is/ are correct?



(i) The ball takes twice as much time to roll from Q to O as it does to roll from P to O.
(ii) The acceleration of the ball at Q is twice as large as the acceleration at P.
(iii) The ball has twice as much K.E. at O when rolling from Q as it does when rolling from P.
(A) i, ii only (B) ii, iii only (C) i only (D) iii only

RD0095

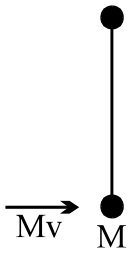
142. The moment of inertia of a solid cylinder about its axis is given by $(1/2)MR^2$. If this cylinder rolls without slipping, the ratio of its rotational kinetic energy to its translational kinetic energy is
(A) 1 : 1 (B) 2 : 2 (C) 1 : 2 (D) 1 : 3

RD0096

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

Angular momentum in rotation + translation

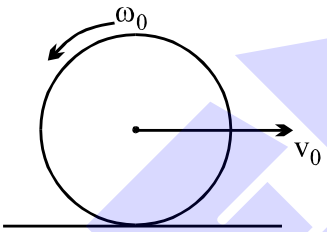
143. Two particles each of mass M are connected by a massless rod of length *l*. The rod is lying on the smooth sufrace. If one of the particle is given an impulse MV as shown in the figure then angular velocity of the rod would be **[IIT-JEE'(Scr)2003]**



(A) v/*l* (B) 2v/*l* (C) v/2*l* (D) None

RD0102

144. A uniform circular disc placed on a rough horizontal surface has initially a velocity v_0 and an angular velocity ω_0 as shown in the figure. The disc comes to rest after moving some distance in the direction of motion. Then $\frac{v_0}{r\omega_0}$ is



(A) $\frac{1}{2}$ (B) 1 (C) $\frac{3}{2}$ (D) 2

RD0097

145. A uniform wheel of radius R is set into rotation about its axis at an angular speed ω. This rotating wheel is now placed on a rough horizontal surface with its axis horizontal. Because of friction at the contact, the wheel accelerates forward and its rotation decelerates till the wheel starts pure rolling on the surface. Find the linear speed of the wheel after it starts pure rolling.

(A) $\frac{\omega R}{3}$ (B) $\frac{\omega R}{6}$ (C) $\frac{2\omega R}{3}$ (D) $\frac{4\omega R}{3}$

RD0377

146. A thin spherical shell lying on a rough horizontal surface is hit by a cue in such a way that the line of action passes through the centre of the shell. As a result, the shell starts moving with a linear speed of the shell after it starts pure rolling on the surface.

(A) $\frac{v}{5}$ (B) $\frac{3v}{5}$ (C) $\frac{4v}{5}$ (D) $\frac{7v}{5}$

RD0378

147. A thin spherical shell of radius R lying on a rough horizontal surface is hit sharply and horizontal by a cue. Where should it be hit so that the shell does not slip on the surface ?

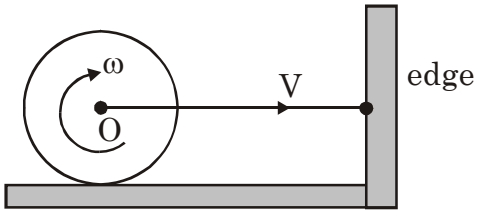
(A) $\frac{2R}{3}$ above the centre (B) $\frac{2R}{3}$ below the centre

(C) $\frac{R}{3}$ above the centre (D) $\frac{R}{3}$ below the centre

RD0379

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, VPE, COM & Coll & Rot. Dy\Eng\04_RD\01_Eng.p65

148. A uniform solid sphere of radius r is rolling on a smooth horizontal surface with velocity v and angular velocity $\omega = (v = \omega r)$. The sphere collides with a sharp edge on the wall as shown in figure. The coefficient of friction between the sphere and the edge $\mu = 1/5$. Just after the collision the angular velocity of the sphere becomes equal to zero. The linear velocity of the sphere just after the collision is equal to



(A) v (B) $\frac{v}{5}$ (C) $\frac{3v}{5}$ (D) $\frac{v}{6}$

RD0380

149. A solid sphere with a velocity (of centre of mass) v and angular velocity ω is gently placed on a rough horizontal surface. The frictional force on the sphere:

(A) must be forward (in direction of v) (B) must be backward (opposite to v)

(C) cannot be zero (D) none of the above **RD0093**

150. A uniform cube with edge a rests on a horizontal plane whose friction coefficient equals k. The cube is set in motion with an initial velocity, travels some distance over the plane and comes to a stand-still.

(A) Angular momentum about any point on ground decreases due to torque of friction.

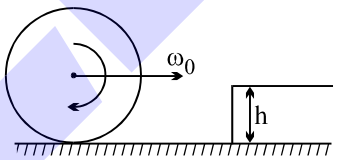
(B) Angular momentum about any point on ground decreases because of combined torque of normal force and gravity.

(C) Angular momentum remains constant

(D) None of these

RD0381

151. A solid uniform sphere of radius R and mass M rolls without slipping with angular velocity ω_0 when it encounters a step of height 0.4 R. Find the angular velocity immediately after inelastic impact with the rough step.

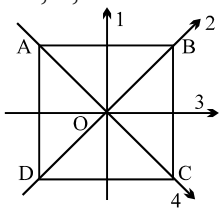


(A) $\frac{5\omega_0}{7}$ **(B)** $\frac{3\omega_0}{7}$ **(C)** $\frac{2\omega_0}{7}$ **(D)** $\frac{\omega_0}{7}$

RD0382

MULTIPLE CORRECT TYPE QUESTIONS

152. ABCD is a square plate with centre O. The moments of inertia of the plate about the perpendicular axis through O is I and about the axes 1, 2, 3 & 4 are I_1, I_2, I_3 & I_4 respectively. It follows that :



(A) $I_2 = I_3$ (B) $I = I_1 + I_4$ (C) $I = I_2 + I_4$ (D) $I_1 = I_3$

RD0103

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Coll & Rot. Dy\Eng\04_RD\01_Eng.p65

153. A rod of weight w is supported by two parallel knifes at end points A and B and is in equilibrium in a horizontal position. The knives are at a distance 'd' from each other. The centre of mass of the rod is at a distance 'x' from A.

(A) the normal reaction at A is $\frac{wx}{d}$ (B) the normal reaction at A is $\frac{w(d-x)}{d}$

(C) the normal reaction at B is $\frac{wx}{d}$ (D) the normal reaction at B is $\frac{w(d-x)}{d}$

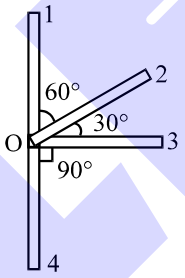
RD0104

154. A block with a square base measuring 'a' × 'a' and height h, is placed on an inclined plane. The coefficient of friction is μ. The angle of inclination (θ) of the plane is gradually increased. The block will

(A) topple before sliding if $\mu > \frac{a}{h}$ (B) topple before sliding if $\mu < \frac{a}{h}$

(C) slide before toppling if $\mu > \frac{a}{h}$ (D) slide before toppling if $\mu < \frac{a}{h}$ **RD0105**

155. A thin rod free to rotate about a horizontal axis through one end O is initially in unstable equilibrium position 1 and falls due to gravity on being displaced infinitesimally. In positions 2,3 and 4 it makes 60°, 90° and 180° with the upward vertical. Let ω_2, ω_3, and ω_4 be angular velocities of the rod in theses positions then



(A) $\omega_4=2\omega_3$ (B) $\omega_4=2\omega_2$ (C) $\omega_3=1.5\omega_2$ (D) $\omega_3=\omega_2\sqrt{2}$

RD0383

156. A thin uniform rigid rod of length *l* is hinged at one end so that it can move in a vertical plane by rotating about a horizontal axis through upper end. The lower end is given a sharp blow and made to acquire a linear velocity v_o. Find the maximum height to which the lower end can rise.



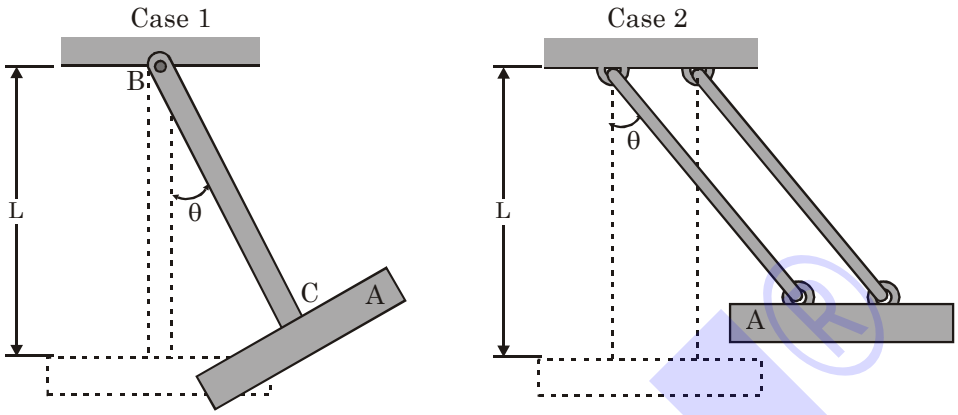
(A) $\frac{v_o^2}{3g}$ for $v_0 < \sqrt{6gl}$ (B) $\frac{v_o^2}{3g}$ for $v_0 \geq \sqrt{6gl}$

(C) 2l for $v_0 \geq \sqrt{6gl}$ (D) 2l for $v_0 < \sqrt{6gl}$

RD0384

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, VPE, COM & Coll & Rot. Dy\Eng\04_RD\01_Eng.p65

157. Slender bar *A* is rigidly connected to a massless rod *BC* in Case 1 and two massless cords in Case 2 as shown. The vertical thickness of bar *A* is negligible compared to *L*. In both cases *A* is released from rest at an angle $\theta = \theta_0$. When $\theta_0 = 0°$



Mark the **CORRECT** statement(s) :-

(A) The kinetic energy will be the same in both the cases.

(B) The speed of the centers of gravity will be same

For Option (C) and (D) : Consider system to be at rest intially in vertical position. If bullet *D* strikes *A* with a speed v_0 initially directed horizontally and becomes embedded in it

(C) The speed of the center of gravity of A immediately after the impact in case-1 will be larger.

(D) The speed of the center of gravity of A immediately after the impact in case-2 will be larger.

RD0385

158. A particle falls freely near the surface of the earth. Consider a fixed point O (not vertically below the particle) on the ground.

(A) Angular momentum of the particle about O is increasing .

(B) Torque of the gravitational force on the particle about O is decreasing.

(C) The moment of inertia of the particle about O is decreasing .

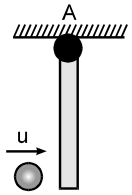
(D) The angular velocity of the particle about O is increasing. **RD0106**

159. A man spinning in free space changes the shape of his body, eg. by spreading his arms or curling up. By doing this, he can change his

(A) moment of inertia (B) angular momentum

(C) angular velocity (D) rotational kinetic energy **RD0107**

160. In the given figure a ball strikes a rod elastically and rod is hinged smoothly at point A. Then which of the statement(s) is/are correct for the collision?



(A) linear momentum of system (ball + rod) is conserved

(B) angular momentum of system about hinged point A is conserved

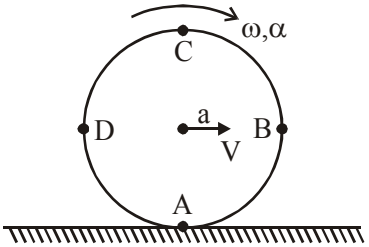
(C) kinetic energy of system is conserved

(D) linear momentum of ball is conserved.

RD0386

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

161. A circular disc of radius R rolls without slipping on a rough horizontal surface. At the instant shown its linear velocity is V, linear acceleration a, angular velocity ω and angular acceleration α. Four points A, B, C and D lie on its circumference such that the diameter AC is vertical & BD horizontal then choose the **CORRECT** option(s).



(A) $V_B = \sqrt{V^2 + (R\omega)^2}$

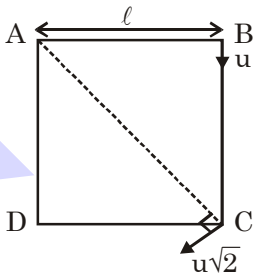
(B) $V_C = V + R\omega$

(C) $a_A = \sqrt{(a - R\alpha)^2 + (\omega^2 R)^2}$

(D) $a_D = \sqrt{(a + \omega^2 R)^2 + (R\alpha)^2}$

RD0108

162. Velocities of some points of a square plate are shown, mass of square plate is 'm'. Choose ***CORRECT*** option(s).



(A) KE of plate is $\frac{mu^2}{3}$.

(B) KE of plate is $\frac{mu^2}{6}$.

(C) Plate is in state of pure rotation about point A.

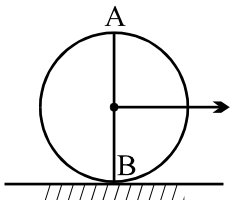
(D) Velocity of centre of plate is $\frac{u}{\sqrt{2}}$.

RD0387

163. A bicycle is in motion. The force of friction exerted by the ground on its wheel is such that it acts:
(A) in backward direction on front wheel and in forward direction on rear wheel when it is accelerating
(B) in forward direction on front wheel and in backward direction on rear wheel when brakes are applied on rear wheel only
(C) in backward direction on front wheel and in forward direction on rear wheel when brakes are applied on rear wheel only
(D) in backward direction on both the wheels when brakes are applied on front wheel

RD0388

164. A uniform disc is rolling on a horizontal surface. At a certain instant B is the point of contact and A is at height 2R from ground, where R is radius of disc.



(A) The magnitude of the angular momentum of the disc about B is thrice that about A.
(B) The angular momentum of the disc about A is anticlockwise.
(C) The angular momentum of the disc about B is clockwise
(D) The angular momentum of the disc about A is equal to that about B.

RD0137

165. A hollow sphere of radius R and mass m is fully filled with non viscous liquid of mass m. It is rolled down a horizontal plane such that its centre of mass moves with a velocity v. If it purely rolls

(A) Kinetic energy of the system is $\frac{5}{6}mv^2$

(B) Kinetic energy of the system is $\frac{4}{5}mv^2$

(C) Angular momentum of the system about a fixed point on ground is $\frac{8}{3}mvR$

(D) Angular momentum of the system about a fixed point on ground is $\frac{14}{5}mvR$

RD0098

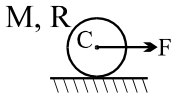
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Coll & Rot. Dy\Eng\04_RD\01_Eng.p65

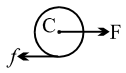
COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 166 to 169

In the following problems, indicate the correct direction of friction force acting on the cylinder, which is pulled on a rough surface by a constant force F.

166. A cylinder of mass M and radius R is pulled horizontally by a force F. The friction force can be given by which of the following diagrams :-



(A) 

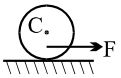
(B) 

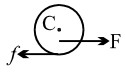
(C) 

(D) cannot be interpreted

RD0109

167. A cylinder is pulled horizontally by a force F acting at a point below the centre of mass of the cylinder, as shown in figure. The friction force can be given by which of the following diagrams :-



(A) 

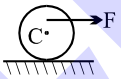
(B) 

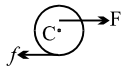
(C) 

(D) cannot be interpreted

RD0109

168. A cylinder is pulled horizontally by a force F acting at a point above the centre of mass of the cylinder, as shown in figure. The friction force can be given by which of the following diagrams



(A) 

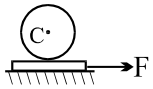
(B) 

(C) 

(D) cannot be interpreted

RD0109

169. A cylinder is placed on a rough plank which in turn is placed on a smooth surface. The plank is pulled with a constant force F. The friction force can be given by which of the following diagrams



(A) 

(B) 

(C) 

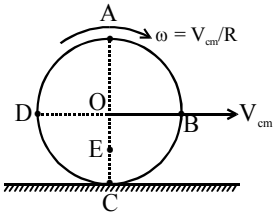
(D) canot be interpreted

RD0109

MATRIX MATCH TYPE QUESTION

170. Consider a body rolling on a horizontal surface as shown in figure (Symbols have their usual meaning)

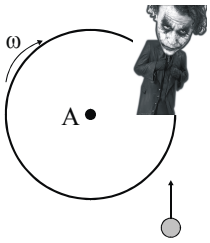
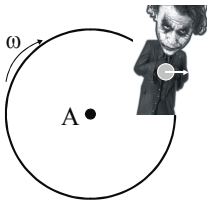
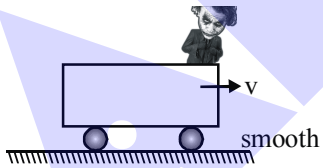
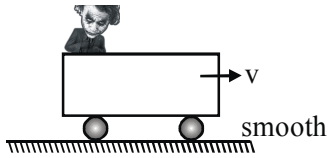
Column-I	Column-II
(A) Velocity is zero	(P) Point A
(B) Speed is maximum	(Q) Point B
(C) $0 <$ speed $< v_{cm}$	(R) Point C
(D) $1.3v_{cm} <$ speed $< 2\ v_{cm}$	(S) Point D
	(T) Point E



RD0111

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

171. Column-I depicts various situations where some sudden events are taking place. Column-II describes changes in various parameters of systems immediately after the events taking place in column-I.

	Column-I	**Column-II**
(A)	Joker is standing on revolving platform and batman throws the ball and joker catches the ball while it was moving horizontally.  Joker, ball and platform is system.	(P) Linear momentum remains conserved.
(B)	Joker throws the ball horizontally and perpendicular to his motion while standing on the revolving platform.  Joker, ball and platform is system.	(Q) Mechanical energy is conserved
(C)	Joker jumps horizontally towards right from the cart which is moving at speed v on smooth horizontal floor.  Joker and cart is the system	(R) Mechanical energy increases.
(D)	Joker drops himself vertically from the moving cart with no horizontal velocity relative to cart.  Joker and cart is the system	(S) Mechanical energy decreases. (T) v or ω changes

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

EXERCISE (O-2)

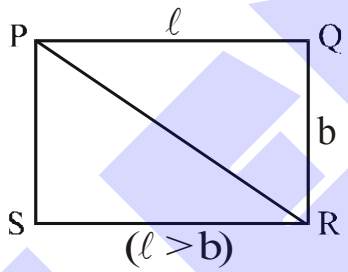
Moment of inertia

1. Let I be the moment of inertia of a uniform square plate about an axis AB that passes through its centre and is parallel to two of its sides. CD is a line in the plane of the plate that passes through the centre of the plate and makes an angle θ with AB. The moment of inertia of the plate about the axis CD is then equal to

(A) I (B) $I\sin^2\theta$ (C) $I\cos^2\theta$ (D) $I\cos^2(\theta/2)$

RD0058

2. Moment of inertia of a rectangular plate about an axis passing through P and perpendicular to the plate is I. Then moment of PQR about an axis perpendicular to the plane of the plate:



(A) about P = I/2 (B) about R = I/2 (C) about P > I/2 (D) about R > I/2

RD0062

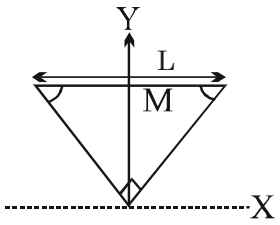
3. The moment of inertia of a solid cylinder about an axis parallel to its length and passing through its centre is equal to its moment of inertia about an axis perpendicular to the length of cylinder and passing through its centre. The ratio of radius of cylinder and its length is

(A) $1:\sqrt{2}$ (B) 1 : 2 (C) $1:\sqrt{3}$ (D) 1 : 3

RD0389

Paragraph for Question No. 4 to 7

The figure shows an isosceles triangular plate of mass M and base L. The angle at the apex is 90°. The apex lies at the origin and the base is parallel to X–axis



4. The moment of inertia of the plate about the z-axis is :-

(A) $\frac{ML^2}{12}$ (B) $\frac{ML^2}{24}$ (C) $\frac{ML^2}{6}$ (D) none of these

RD0143

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

5. The moment of inertia of the plate about the x-axis is :-

(A) $\frac{ML^2}{8}$ (B) $\frac{ML^2}{32}$ (C) $\frac{ML^2}{24}$ (D) $\frac{ML^2}{6}$

RD0144

6. The moment of inertia of the plate about its base parallel to the x-axis is :-

(A) $\frac{ML^2}{18}$ (B) $\frac{ML^2}{36}$ (C) $\frac{ML^2}{24}$ (D) none of these

RD0143

7. The moment of inertia of the plate about the y-axis is :-

(A) $\frac{ML^2}{6}$ (B) $\frac{ML^2}{8}$ (C) $\frac{ML^2}{24}$ (D) none of these

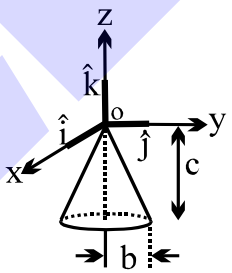
RD0143

8. A body is in equilibrium under the influence of a number of forces. Each force has a different line of action. The minimum number of forces required is
(A) 2, if their lines of action pass through the centre of mass of the body.
(B) 3, if their lines of action are not parallel.
(C) 3, if their lines of action are parallel.
(D) 4, if their lines of action are parallel and all the forces have the same magnitude.

RD0132

Torque and equilibrium

9. A solid cone hangs from a frictionless pivot at the origin O, as shown. If $\hat{i}$, $\hat{j}$ and $\hat{k}$ are unit vectors, and a, b, and c are positive constants, which of the following forces F applied to the rim of the cone at a point P results in a torque τ on the cone with a negative component τ_z?



(A) $F = a\hat{k}$, P is (0,b,–c) (B) $F = -a\hat{k}$, P is (0,–b,–c)
(C) $F = a\hat{j}$, P is (–b,0,–c) (D) None

RD0113

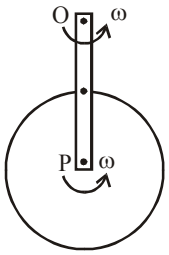
10. A heavy seesaw (i.e., not massless) is out of balance. A light girl sits on the end that is tilted downward, and a heavy body sits on the other side so that the seesaw now balances. If they both move forward so that they are one-half their original distance from the pivot point (the fulcrum) what will happen to the seesaw ?
(A) The side the body is sitting on will tilt downward
(B) The side the girl is sitting on will once again tilt downward
(C) Nothing ; the seesaw will still be balanced
(D) It is impossible to say without knowing the masses and the distances

RD0114

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, VPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

Fixed axis

11. A rod of mass m and length 2R can rotate about an axis passing through O in vertical plane. A disc of mass m and radius R is hinged to the other end P of the rod and can freely rotate about P. When disc is at lowest point both rod and disc has angular velocity ω. If rod rotates by maximum angle θ = 60° with downward vertical, the ω in terms of R and g will be (all hinges are smooth) :-



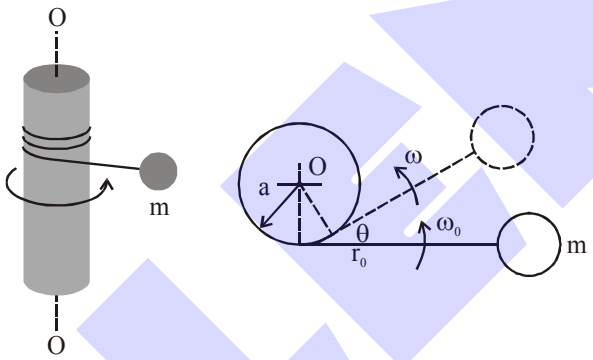
(A) $\sqrt{\frac{9g}{16R}}$ (B) $\sqrt{\frac{3g}{23R}}$

(C) $\frac{1}{3}\sqrt{\frac{g}{R}}$ (D) None of these

RD0390

Angular momentum & its conservation

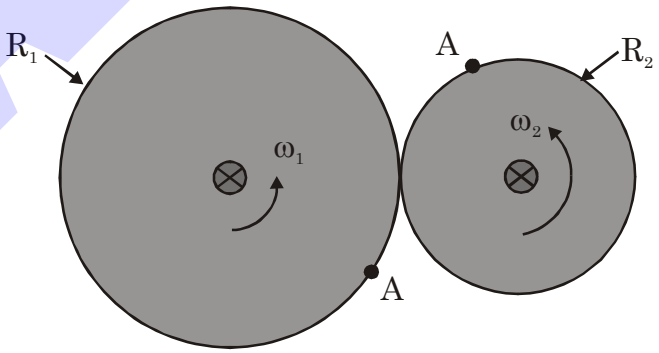
12. The small particle of mass m is given an initial high velocity 10 m/s in the horizontal plane and winds its cord around the fixed vertical shaft of radius 1cm. All motion occurs essentially in the horizontal plane.



(A) Tension does not do any work.
(B) Speed remains constant.
(C) Angular momentum of the particle about axis of cylinder is constant.
(D) If length of thread is 10 m, particle strikes cylinder after 500 sec.

RD0391

13. Consider two discs rotating about their symmetry axes as shown. They have different radii and different angular speeds. As time progresses, due to the friction at the contact points, their angular speeds change. About which point can their angular momentum be conserved?

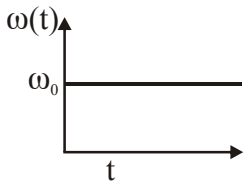


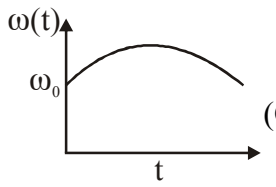
(A) Point of contact
(B) Centre of the left disc
(C) Centre of the right disc
(D) The angular momentum of system is not conserved about any point.

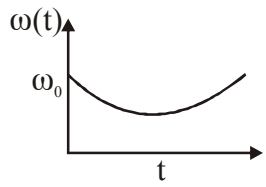
RD0392

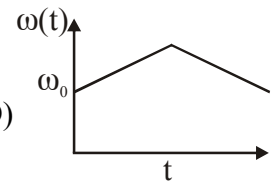
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

14. A circular platform is free to rotate in a horizontal plane about a vertical axis passing through its centre. A tortoise is sitting at the edge of the platform. Now, the platform is given an angular velocity ω_0. When the tortoise move along a chord of the platform with a constant velocity (with respect to the platform), the angular velocity of the platform ω(t). will vary with time t as **[IIT-JEE 2002]**

(A) 

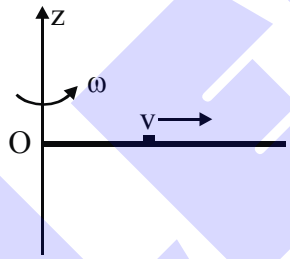
(B) 

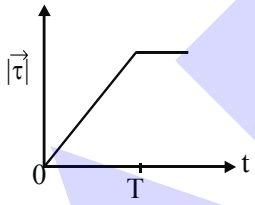
(C) 

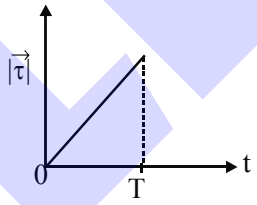
(D) 

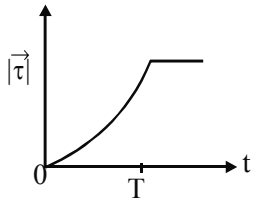
RD0085

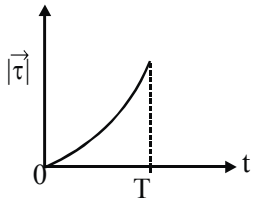
15. A thin uniform rod, pivoted at OP, is rotating in the horizontal plane with constant angular speed ω. as shown in the figure. At time t = 0, a small insect starts from O and moves with constant speed v with respect to the rod towards the other end. If reaches the end of the rod at t = T and stops. The angular speed of the system remains ω throughout. The magnitude of the torque $(|\tau|)$ on the system about O, as a function of time is best represented by which plot? **[IIT-JEE 2012]**



(A) 

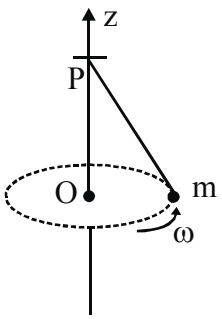
(B) 

(C) 

(D) 

RD0174

16. A small mass m is attached to a massless string whose other end is fixed at P as shown in the figure. The mass is undergoing circular motion in the x-y plane with centre at O and constant angular speed ω. If the angular momentum of the system, calculated about O and P are denoted by $\vec{L}_O$ and $\vec{L}_P$ respectively, then **[IIT-JEE 2012]**



(A) $\vec{L}_O$ and $\vec{L}_P$ do not vary with time

(B) $\vec{L}_O$ varies with time while $\vec{L}_P$ remains constant

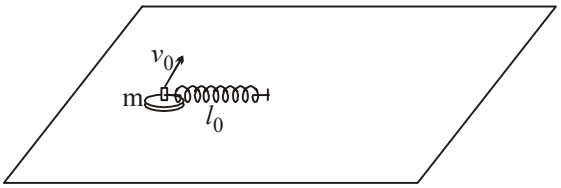
(C) $\vec{L}_O$ remains constant while $\vec{L}_P$ varies with time

(D) $\left|\vec{L}_O\right|$ and $\left|\vec{L}_P\right|$ both do not vary with time **RD0175**

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, VPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

Paragraph for Question No. 17 to 20

A spring having initial unstretched length ℓ_0 is lying on a smooth table. It's one end is fixed and the other one is fastened to a small particle of mass m. The particle is imparted an initial speed v_0 horizontally in a direction perpendicular to the spring. In the course of the motion in horizontal plane, the maximum elongation of the spring is $\Delta\ell = \ell_0/10$. (Given : m = 0.1 kg, ℓ_0 = 1m, v_0 = 11 m/s).



17. In the course of motion, which of the quantities relating to spring block system are conserved ?
(A) kinetic energy (B) momentum (C) angular momentum (D) potential energy

RD0147

18. Which of the following is correct about initial situation and situation at maximum elogation ?
(A) the orientiation of spring in both positions should be perpendicular to each other.
(B) the velocity at maximum extension should be zero
(C) the velocity at maximum extension as well as at initial position should be perpendicular to spring.
(D) Acceeration should be zero at the maximum extension as well as at initial position.

RD0147

19. **Student-A :** at maximum extension $mv'(\ell_0 + \Delta\ell) = mv_0\ell_0$.

Student-B : at maximum extension $k\Delta\ell = \dfrac{mv'^2}{(\ell_0 + \Delta\ell)}$. **Student-C :** $\dfrac{1}{2}mv_0^2 = \dfrac{1}{2}mv'^2 + \dfrac{1}{2}k\Delta\ell^2$.

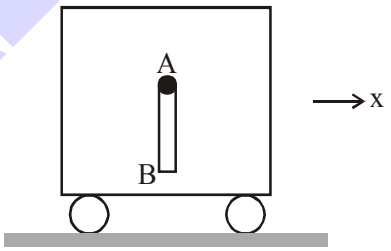
where v' is velocity at instant of maximum extension :
(A) Only Student-A and B are correct (B) Only Student-A and C are correct
(C) Only Student-B and C are correct (D) All are correct

RD0147

20. The value of spring constant (in N/m) is :
(A) 100 (B) 210 (C) 420 (D) 105

RD0147

21. A uniform rod AB of length ℓ and mass M hangs from point A at which it is freely hinged in a car moving with velocity v_0. The rod can rotate in vertical plane about the axis at A. If the car suddenly stops,



(A) The angular speed ω with which the rod starts rotating is $\dfrac{3v_0}{2\ell}$.

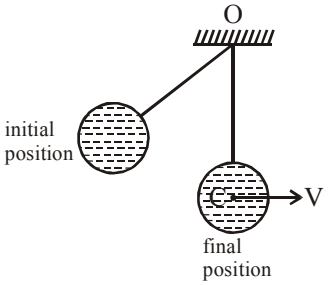
(B) The minimum value of v_0 so that the rod completes the rotation $\sqrt{\dfrac{8}{3}g\ell}$.

(C) Loss of energy during the process $\dfrac{1}{8}Mv_0^2$.

(D) There is no loss of energy. **RD0142**

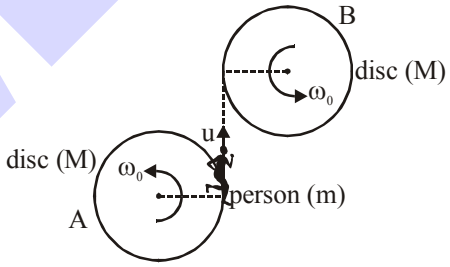
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

22. A massles rod has a massless hollow sphere attached to it. This sphere can be fully filled either with a liquid (non viscous) or with a solid (rigidly fitted into sphere) of same mass. System is released from rest from initial position (as shown). When it reaches final position which of the following is/are true for the system.



(A) Kinetic energy in case of liquid will be more than in case of solid.
(B) Velocity of centre (V) in case of liquid will be more than in case of solid.
(C) About C angular momentum in case of liquid will be more than in case of solid.
(D) About C angular momentum in case of liquid will be less than in case of solid. **RD0141**

23. Two identical discs each of mass M(= 4m) and radius R are rotating in opposite sense with equal angular speed ω_0 about vertical axes passing through their centres, (as shown in figure). A person of mass m sitting on circumference of disc A jumps with a tangential relative velocity u (after jumping) w.r.t one rotating disc (A) and lands on other disc (B) also tangential. Now the second disc (B) comes to a stop. Find the relative velocity u.



(A) $u = \frac{\omega_0 R}{2}$

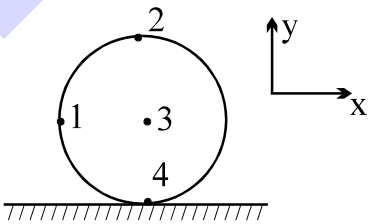
(B) $u = \omega_0 R$

(C) $u = \frac{3\omega_0 R}{2}$

(D) $u = 2\omega_0 R$ **RD0115**

Rolling kinematics

24. A rigid cylinder is kept on a smooth horizontal surface as shown. If **Column-I** indicates velocities of various points (3-centre of cylinder, 2- top point, 4-bottom point, 1- on the level of 3 at the rim) on it shown, choose correct state of motion from **Column-II**.



	Column-I		**Column-II**
(A)	$\vec{v}_1 = \hat{i} + \hat{j}, \vec{v}_2 = 2\hat{i}$	(P)	Pure rotation about centre
(B)	$\vec{v}_1 = \hat{i} + \hat{j}, \vec{v}_3 = -\hat{i}$	(Q)	Rolling without slipping to left
(C)	$\vec{v}_2 = \hat{i}, \vec{v}_3 = 0$	(R)	Rolling without slipping to right
(D)	$\vec{v}_4 = 0, \vec{v}_1 = -\hat{i} - \hat{j}$	(S)	Not possible

RD0152

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, VPE, COM & Coll & Rot. Dy\Eng\04_RD\01_Eng.p65

25. A ring rolls without slipping on the ground. Its centre C moves with a constant speed u. P is any point on the ring. The speed of P with respect to the ground is v.

(A) $0 \le v \le 2u$

(B) $v = u$, if CP is horizontal

(C) $v = u$, if CP makes an angle of 30° with the horizontal and P is below the horizontal level of C.

(D) $v = \sqrt{2}u$, if CP is horizontal

RD0139

26. A wheel of radius r rolling on a straight line, the velocity of its centre being v. At a certain instant the point of contact of the wheel with the grounds is M and N is the highest point on the wheel (diametrically opposite to M). The **INCORRECT** statement is:

(A) The velocity of any point P of the wheel is proportional to MP.

(B) Points of the wheel moving with velocity greater than v form a larger area of the wheel than points moving with velocity less than v.

(C) The point of contact M is instantaneously at rest.

(D) The velocities of any two parts of the wheel which are equidistant from centre are equal.

RD0127

27. A straight rod of length L is released on a frictionless horizontal floor in a vertical position. As it falls + slips, the distance of a point on the rod from the lower end, which follows a quarter circular locus is :-

(A) L/2 (B) L/4 (C) L/8 (D) None

RD0119

28. A ladder of length L is slipping with its ends against a vertical wall and a horizontal floor. At a certain moment, the speed of the end in contact with the horizontal floor is v and the ladder makes an angle $\alpha = 30°$ with the horizontal. Then the speed of the ladder's center must be :-

(A) $2v/\sqrt{3}$ (B) v/2 (C) v (D) None

RD0128

29. In the previous question, if $dv/dt = 0$, then the angular acceleration of the ladder when $\alpha = 45°$ is :-

(A) $2v^2/L^2$ (B) $v^2/2L^2$ (C) $\sqrt{2}[v^2/L^2]$ (D) None

RD0129

30. A ring of radius R is rolling purely on the outer surface of a pipe of radius 4R. At some instant, the center of the ring has a constant speed = v. Then ,

(A) The acceleration of the point on the ring which is in contact with the surface of the pipe is $4v^2/5R$

(B) The acceleration of the point on the ring which is in contact with the surface of the pipe is $3v^2/5R$

(C) The acceleration of the point on the ring which is farthest from the center of the pipe, at the given moment is $6v^2/5R$

(D) The acceleration of the point on the ring which is farthest from the center of the pipe, at the given moment is $3v^2/5R$

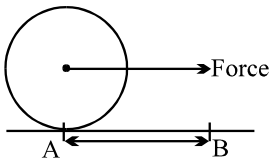
RD0393

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

Rolling dynamics

31. If a cylinder is rolling down the incline with sliding.
(A) after some time it may start pure rolling
(B) after sometime it will start pure rolling
(C) it may be possible that it will never start pure rolling
(D) none of these **RD0135**

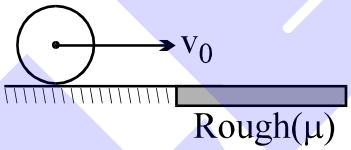
32. A disc of circumference s is at rest at a point A on a horizontal surface when a constant horizontal force begins to act on its centre. Between A and B there is sufficient friction to prevent slipping, and the surface is smooth to the right of B. AB = s. The disc moves from A to B in time T. To the right of B,



(A) the angular acceleration of the disc will disappear, linear acceleration will remain unchanged
(B) linear acceleration of the disc will increase
(C) the disc will make one rotation in time T/2
(D) the centre of mass of disc will cover a distance greater than s in further time T. **RD0138**

Paragraph for Question No. 33 to 34

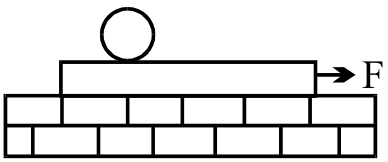
A ring of mass M and radius R sliding with a velocity v_0 suddenly enters into rough surface where the coefficient of friction is μ, as shown in figure.



33. Choose the correct statement(s)
(A) As the ring enters on the rough surface, the limiting friction force acts on it
(B) The direction of friction is opposite to the direction of motion
(C) The friction force accelerates the ring in the clockwise sense about its centre of mass
(D) As the ring enters on the rough surface it starts rolling **RD0150**

34. Choose the correct statement(s)
(A) The momentum of the ring is conserved
(B) The angular momentum of the ring is conserved about its centre of mass
(C) The angular momentum of the ring conserved about any point on the horizontal surface in line of friction.
(D) The mechanical energy of the ring is conserved **RD0150**

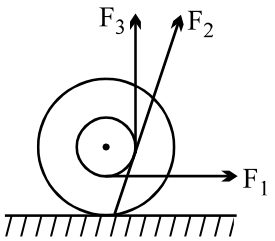
35. A plank with a uniform sphere placed on it rests on a smooth horizontal plane. Plank is pulled to right by a constant force F. If sphere does not slip over the plank. Which of the following is correct.



(A) Acceleration of the centre of sphere is less than that of the plank.
(B) Work done by friction acting on the sphere is equal to its total kinetic energy.
(C) Total kinetic energy of the system is equal to work done by the force F
(D) None of the above **RD0136**

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, VPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

36. A yo-yo is resting on a perfectly rough horizontal table. Forces F_1, F_2 and F_3 are applied separately as shown. The correct statement is :-



(A) when F_3 is applied the centre of mass will move to the right.

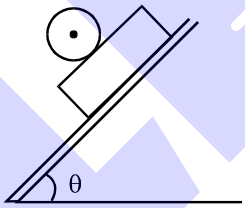
(B) when F_2 is applied the centre of mass will move to the left.

(C) when F_1 is applied the centre of mass will move to the right.

(D) when F_2 is applied the centre of mass will move to the right.

RD0125

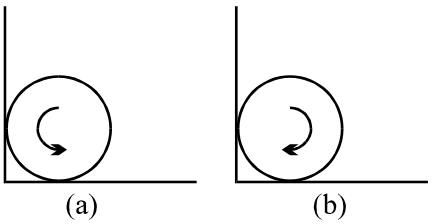
37. A plank of mass M is placed over smooth inclined plane and a sphere is also placed over the plank. Friction is sufficient between sphere and plank. If plank and sphere are released from rest, the frictional force on sphere is:



(A) up the plane (B) down the plane (C) horizontal (D) zero

RD0122

38. A sphere is placed rotating with its centre initially at rest in a corner as shown in figure (a) & (b). Coefficient of friction between all surfaces and the sphere is 1/3. Find the ratio of the frictional force $\frac{f_a}{f_b}$ by ground in situations (a) & (b).



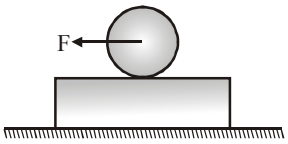
(A) 1 (B) $\frac{9}{10}$ (C) $\frac{10}{9}$ (D) none

RD0120

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Coll & Rot. Dy\Eng\04_RD\01_Eng.p65

Paragraph for Question No. 39 to 40

A disc of mass m and radius R is placed over a plank of same mass m. There is sufficient friction between disc and plank to prevent slipping. A force F is applied at the centre of the disc. Friction is absent between plank and ground.



39. Acceleration of the plank is :-

(A) $\frac{F}{2m}$ (B) $\frac{3F}{4m}$ (C) $\frac{F}{4m}$ (D) $\frac{3F}{2m}$

RD0148

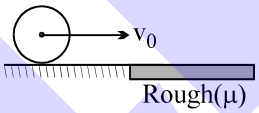
40. Force of friction between the disc and the plank is :-

(A) $\frac{F}{2}$ (B) $\frac{F}{4}$ (C) $\frac{F}{3}$ (D) $\frac{2F}{3}$

RD0148

Paragraph for Question No. 41 to 42

A ring of mass M and radius R sliding with a velocity v_0 suddenly enters into rough surface where the coefficient of friction is μ, as shown in figure.



41. Choose the correct statement(s)

(A) The ring starts its rolling motion when the centre of mas stationary.

(B) The ring starts rolling motion when the point of contact becomes stationary.

(C) The time after which the ring starts rolling is $\frac{v_0}{2\mu g}$.

(D) The rolling velocity is $\frac{v_0}{2}$. **RD0150**

42. Choose the correct alternative(s) :-

(A) The linear distance moved by the centre of mass before the ring starts rolling is $\frac{3v_0^2}{8\mu g}$

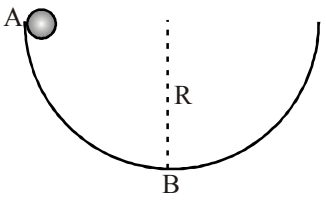
(B) The net work done by friction force is $-\frac{3}{8}mv_0^2$

(C) The loss is kinetic energy of the ring is $\frac{mv_0^2}{4}$

(D) The gain in rotational kinetic energy is $+\frac{mv_0^2}{8}$ **RD0150**

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, VPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

43. A small sphere A of mass m and radius r rolls without slipping inside a large fixed hemispherical bowl of radius R (>> r) as shown in figure. If the sphere starts from rest at the top point of the hemisphere find the normal force exerted by the small sphere on the hemisphere when it is at the bottom B of the hemisphere.

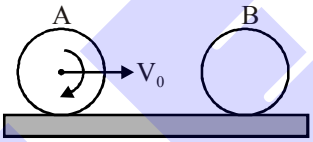


(A) $\frac{10}{7}$mg (B) $\frac{17}{7}$mg (C) $\frac{5}{7}$ mg (D) $\frac{7}{5}$mg

RD0121

Angular momentum in translation + rotation

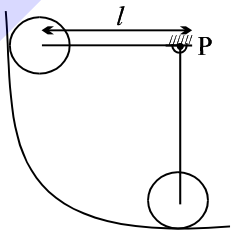
44. A hollow smooth uniform sphere A of mass 'm' rolls without sliding on a smooth horizontal surface. It collides head on elastically with another stationary smooth solid sphere B of the same mass m and same radius. The ratio of kinetic energy of 'B' to that of 'A' just after the collision is :



(A) 1 : 1 (B) 2 : 3 (C) 3 : 2 (D) None

RD0126

45. A sphere of mass M and radius R is attached by a light rod of length ℓ to a point P. The sphere rolls without slipping on a circular track as shown. It is released from the horizontal position. The angular momentum of the system about P when the rod becomes vertical is :



(A) $M\sqrt{\frac{10}{7}gl}\,[l+R]$ (B) $M\sqrt{\frac{10}{7}gl}\left[l+\frac{2}{5}R\right]$

(C) $M\sqrt{\frac{10}{7}gl}\left[l+\frac{7}{5}R\right]$ (D) $M\sqrt{\frac{10}{7}gl}\left[l-\frac{2}{5}R\right]$

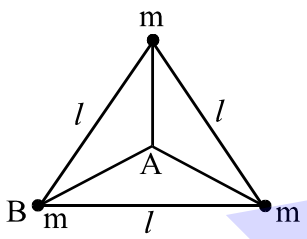
RD0099

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Coll & Rot. Dy\Eng\04_RD\01_Eng.p65

EXERCISE (S)

Moment of inertia

1. Three equal masses m are rigidly connected to each other by massless rods of length l forming an equilateral triangle, as shown in the figure. What is the ratio of the moment of inertia of the assembly for an axis through B compared with that for an axis through A (centroid). Both the axis are perpendicular to the plane of triangle.

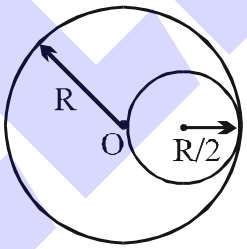


RD0001

2. Two rods of equal mass m and length l lie along the x axis and y axis with their centres at origin. What is the moment of inertia of the system about the line x=y :

RD0002

3. A hole of radius R/2 is cut from a solid sphere of radius R. If the mass of the remaining part of sphere is M , then find moment of inertia of the body about an axis through O.



RD0029

4. Find the MI of a rod about an axis through its centre of mass and perpendicular to the length whose linear density varies as $\lambda = ax$ where a is a constant and x is the position of an element of the rod relative to its left end. The length of the rod is ℓ.

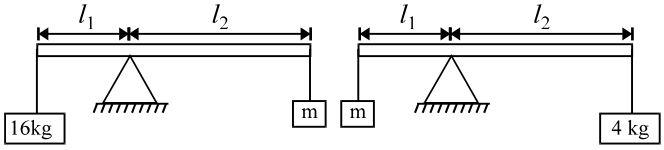
RD0030

5. The surface density (mass/area) of a circular disc of radius a depends on the distance from the centre as $\rho(r) = A + Br$. Find then moment of inertia about the line perpendicular to the plane of the disc through its centre.

RD0394

Rotational equilibrium & toppling

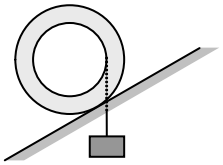
6. In an experiment with a massless beam balance an unknown mass m is balanced by two known masses of 16kg and 4 kg as shown in figure.Find the value of the unknown mass m.



RD0004

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Collision & Rot. Dy\Eng\04_RD\01_Eng.p65

7. A 3.0 kg bobbin consists of a central cylinder of radius 5.0 cm and two end plates each of radius 6.0 cm. It is placed on a slotted incline, where friction is sufficient to prevent sliding. A block of mass 4.5 kg is suspended from a cord wound around the bobbin and passing through the slot under the incline. If the bobbin is in static equilibrium, what is the angle of tilt of the incline?



RD0006

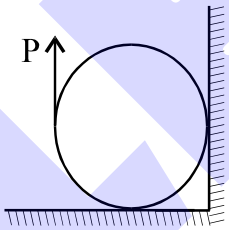
8. A uniform ladder of mass 10 kg leans against a smooth vertical wall making an angle of 53° with it. The other end rests on a rough horizontal floor. Find the normal force and the frictional force that the floor exerts on the ladder.

RD0395

9. A 6.5 m long ladder rests against a vertical wall reaching a height of 6.0 m. A 60 kg man stands half way up the ladder.
(a) Find the torque of the force exerted by the man on the ladder about the upper end of the ladder.
(b) Assuming the weight of the ladder to be negligible as compared to the man and assuming the wall to be smooth, find the force exerted by the ground on the ladder.

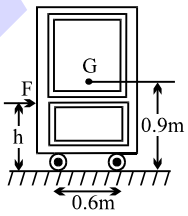
RD0396

10. Figure shows a vertical force applied tangentially to a uniform cylinder of weight F_g. The coefficient of static friction between the cylinder and both surfaces is 0.500. In terms of F_g, find the maximum force **P** that can be applied that does not cause the cylinder to rotate.



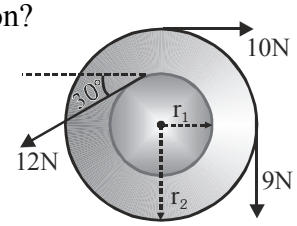
RD0031

11. A 20 kg cabinet is mounted on small casters that allow it to move freely ($\mu = 0$) on the floor. If a 100 N force is applied as shown, determine



(a) the acceleration of the cabinet,
(b) the range of values of h for which the cabinet will not tip. **RD0034**

12. In the following figure r_1 and r_2 are 5 cm and 30 cm respectively. If the moment of inertia of the wheel is 5100 kg-m^2 about the axis passing through its centre and perpendicular to the plane of wheel, then what will be its angular acceleration?

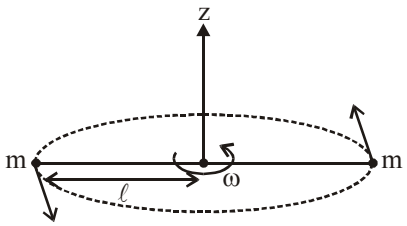


RD0007

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Coll & Rot. Dy\Eng\04_RD\01_Eng.p65

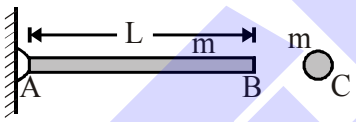
Fixed axis

13. A dumbbell consists of two identical particles of mass m connected by a rigid light rod of length 2ℓ. The dumbbell is set spinning with angular speed ω_0 on a surface with a small friction coefficient μ_k. If dumbbell stops in time $t = \dfrac{K\omega_0\ell}{2\mu g}$ where K is a constant, then find the value of K.



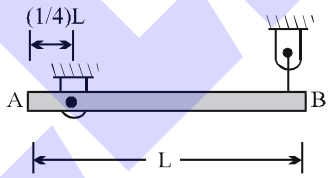
RD0008

14. A uniform bar AB of mass m and a ball of the same mass are released from rest from the same horizontal position. The bar is hinged at end A. There is gravity downwards. What is the distance of the point from point B that has the same acceleration as that of ball, immediately after release?



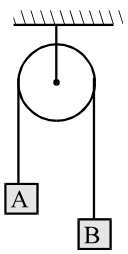
RD0009

15. A uniform beam of length L and mass m is supported as shown. If the cable at B suddenly breaks, determine; (a) the acceleration of end B. (b) the reaction at the pin support.



RD0010

16. In the figure, A & B are two blocks of mass 4 kg & 2 kg respectively attached to the two ends of a light string passing over a disc C of mass 40 kg and radius 0.1 m. The disc is free to rotate about a fixed horizontal axes, coinciding with its own axis. The system is released from rest and the string does not slip over the disc. Find: (i) the linear acceleration of mass B .

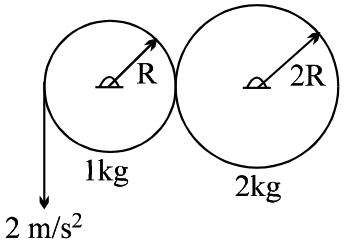


(ii) the number of revolutions made by the disc at the end of 10sec. from the start.
(iii) the tension in the string segment supporting the block A.

RD0011

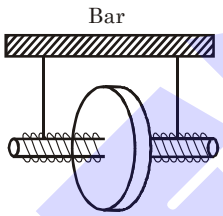
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

17. Two discs A and B touch each other as shown in figure. A rope tightly wound on A is pulled down at 2 m/s^2. Find the friction force between A and B if slipping is absent



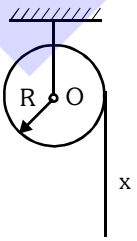
RD0012

18. The ends of thin threads tightly wound on the axle of radius r of the Maxwell disc as shown in figure are attached to a horizontal bar. When the disc unwinds, the bar is raised to keep the disc at the same height. The mass of the disc with the axle is equal to m, the moment of inertia of the arrangement relative to its axis is I. Find the tension of each thread and the acceleration of the bar.



RD0397

19. A uniform cylinder of radius R and mass M can rotate freely about a stationary horizontal axis O. A thin cord of length ℓ and mass m is wound on the cylinder in a single layer. Find the angular acceleration of the cylinder as a function of the length x of the hanging part of the cord. The wound part of the cord is supposed to have its centre of gravity on the cylinder axis.



RD0398

20. A string is wrapped around the rim of a wheel of moment of inertia 0.20 kg-m^2 and radius 20 cm. The wheel is free to rotate about its axis. Initially, the wheel is at rest. The string is now pulled by a force of 20 N. Find the angular velocity of the wheel after 5.0 seconds.

RD0399

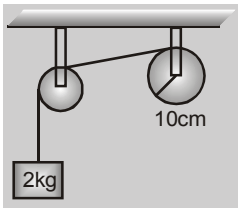
21. A light rod of length 1 m is pivoted at its centre and two masses of 5 kg and 2 kg are hung from the ends as shown in figure. Find the initial angular acceleration of the rod assuming that it was horizontal in the beginning.



RD0400

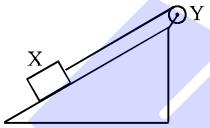
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

22. A string is wrapped on a wheel of moment of inertia 0.20 kg-m^2 and radius 10 cm and goes through a light pulley to support a block of mass 2.0 kg as shown in figure. Find the acceleration of the block.



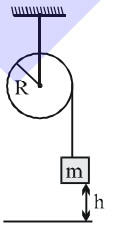
RD0401

23. A block X of mass 0.5kg is held by a long massless string on a frictionless fixed inclined plane of inclination 30° to the horizontal. The string is wound on a uniform solid cylindrical drum Y of mass 2kg and of radius 0.2 m as shown in the fig. The drum is given an initial angular velocity such that the block X starts moving up the plane. (g = 9.8 m/s^2)
(i) Find the tension in the string during the motion
(ii) At a certain instant of time the magnitude of the angular velocity of Y is 10 rad/sec. Calculate the distance travelled by X from that instant of time until it comes to rest.



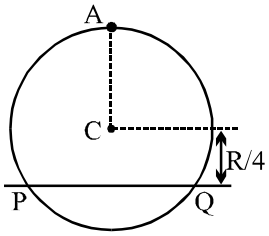
RD0033

24. A mass m is attached to a pulley through a cord as shown in the figure. The pulley is a solid disk with radius R. The cord does not slip on the disk. The mass is released from rest at a height h from the ground and at the instant the mass reaches the ground, the disk is rotating with angular velocity ω. Find the mass of the disk.



RD0013

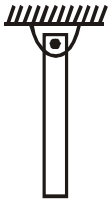
25. A uniform circular disc has radius R and mass m. A particle also of mass m is fixed at a point A on the edge of the disc as in figure. The disc can rotate freely about a fixed horizontal chord PQ that is at a distance R/4 from the centre C of the disc. The line AC is perpendicular to PQ. Initially the disc is held vertical with the point A at its highest position. It is then allowed to fall so that it starts rotating about PQ. Find the linear speed of the particle at it reaches its lowest position.



RD0014

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

26. A rod of mass m & length ℓ is hinged at it's upper end. It can rotate in vertical plane. It is given angular velocity ω so that it can complete vertical circle. Find (a) ω (b) Tension at centre of rod at initial moment.



RD0015

27. A uniform rod of mass m and length 1 is kept vertical with the lower end clamped. It is slightly pushed to let it fall down under gravity. Find its angular speed when the rod is passing through its lowest position . Neglect any friction at the clamp. What will be the linear speed of the free end at this instant ?

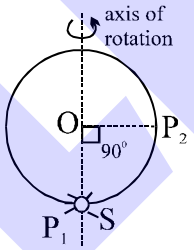
RD0402

28. The door of an automobile is open and perpendicular to the body. The automobile starts with an acceleration of 2 ft/sec^2, and the width of the door is 30 inches. Treat the door as a uniform rectangle, and neglect friction to find the speed of its outside edge as seen by the driver when the door closes.

RD0036

Angular momentum & its conservation

29. A uniform ring is rotating about vertical axis with angular velocity ω initially. A point insect (S) having the same mass as that of the ring starts walking from the lowest point P_1 and finally reaches the point P_2(as shown in figure). What is the final angular velocity of the ring?

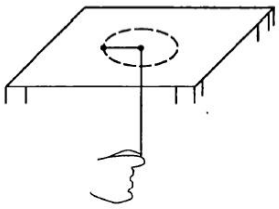


RD0016

30. Two men, each of mass 75 kg , stand on the rim of a horizontal large disc, diametrically opposite to each other. The disc has a mass 450 kg and is free to rotate about its axis. Each man simultaneously start along the rim clockwise with the same speed and reaches their original starting points on the disc. Find the angle turned by the disc with respect to the ground in this duration.

RD0017

31. Figure shows a mass m placed on a frictionless horizontal table and attached to a string passing through a small hole in the surface. Initially, the mass moves in a circle of radius r_0 with a speed v_0 and the free end of the string is held by a person. The person pulls on the string slowly to decrease the radius of the circle to r. (a) Find the tension in the string when the mass moves in the circle of radius r. (b) Calculate the change-in the kinetic energy of the mass.



RD0403

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Collision & Rot. Dy\Eng\04_RD\01_Eng.p65

32. A particle moves along a closed trajectory in a central field of force where the particle's potential energy $U = kr^2$ (k is a positive constant, r is the distance of the particle from the centre O of the field). Find the mass of the particle if its minimum distance from the point O equals r_1 and its velocity at the point farthest from O equals v_2.

RD0404

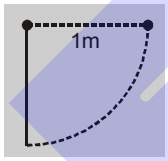
33. Two horizontal discs rotate freely about a vertical axis passing through their centres. The moments of inertia of the discs relative to the axis are equal to I_1 and I_2, and the angular velocities to ω_1 and ω_2. When the upper disc fell on the lower one, both discs began rotating, after some time, as a single whole (due to friction). Find:

(a) the steady-state angular rotation velocity of the discs;

(b) the work performed by the friction forces in this process.

RD0405

34. A metre stick weighing 240 g is pivoted at its upper end in such a way that it can freely rotate in a vertical plane through this end (figure). A particle of mass 100 g is attached to the upper end of the stick through a light string of length 1 m. Initially, the rod is kept vertical and the string horizontal when the system is released from rest. The particle collides with the lower end of the stick and sticks there. Find the maximum angle through which the stick will rise.



RD0406

35. A circular disc of mass 300 gm and radius 20 cm can rotate freely about a vertical axis passing through its centre O. A small insect of mass 100 gm is initially at a point A on the disc (which is initially stationary) the insect starts walking from rest along the rim of the disc with such a time varying relative velocity that the disc rotates in the opposite direction with a constant angular acceleration $= 2\pi$ rad/s^2. After some time T, the insect is back at the point A. By what angle θ has the disc rotated till now ; as seen by a stationary earth observer? Also find the time T.

RD0038

36. A uniform disc of mass m and radius R rotates about a fixed vertical axis passing through its centre with angular velocity ω. A particle of same mass m and having velocity $2\omega R$ towards centre of the disc collides with the disc moving horizontally and sticks to its rim. Find

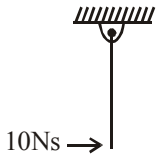
(a) the angular velocity of the disc.

(b) the impulse on the particle due to disc.

(c) the impulse on the disc due to hinge.

RD0049

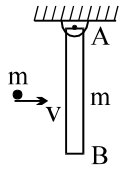
37. A thin uniform straight rod of mass 2 kg and length 1 m is free to rotate about its upper end when at rest. It receives an impulsive blow of 10 Ns at its lowest point, normal to its length as shown in figure. Find the kinetic energy of rod just after impact.



RD0018

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, VPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

38. A uniform rod AB of length L and mass m is suspended freely at A and hangs vertically at rest when a particle of same mass m is fired horizontally with speed v to strike the rod at its mid point. If the particle comes to rest after the impact, then find the impulsive reaction at A.



RD0027

39. Figure shows two cylinders of radii r_1 and r_2 having moments of inertia I_1 and I_2 about their respective axes. Initially, the cylinders rotate about their axes with angular speeds ω_1 and ω_2 as shown in the figure. The cylinders are moved closer to touch each other keeping the axes parallel. The cylinders first slip over each other at the contact but the slipping finally ceases due to the friction between them. Find the angular speeds of the cylinders after the slipping ceases can we conserve angular momentum of system about any point ?



RD0407

40. A flywheel of moment of inertia 5.0 kg-m^2 is rotated at a speed of 60 rad/s. Because of the friction at the axle it comes to rest in 5.0 minutes. Find

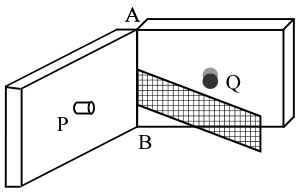
(a) the average torque of the friction.

(b) the total work done by the friction and

(c) the angular momentum of the wheel 1 minute before it stops rotating.

RD0408

41. Two heavy metallic plates are joined together at 90^0 to each other. A laminar sheet of mass 30 Kg is hinged at the line AB joining the two heavy metallic plates. The hinges are frictionless. The moment of inertia of the laminar sheet about an axis parallel to AB and passing through its centre of mass is 1.2 Kg-m^2. Two rubber obstacles P and Q are fixed, one on each metallic plate at a distance 0.5 m from the line AB. This distance is chosen so that the reaction due to the hinges on the laminar sheet is zero during the impact. Initially the laminar sheet hits one of the obstacles with an angular velocity 1 rad/s and turns back. If the impulse on the sheet due to each obstacle is 6 N–s. **[IIT-JEE' 2001]**



(a) Find the location of the centre of mass of the laminar sheet from AB.

(b) At what angular velocity does the laminar sheet come back after the first impact ?

(c) After how many impacts, does the laminar sheet come to rest ?

RD0051

42. A ball of mass m is thrown at an angle α to the horizontal with the initial velocity v_0, Find the time dependence of the magnitude of the ball's angular momentum vector relative to the point from which the ball is thrown. Find the angular momentum M at the highest point of the trajectory if m = 130 g, α = 45°, and v_0 = 25 m/s. The air drag is to be neglected.

RD0409

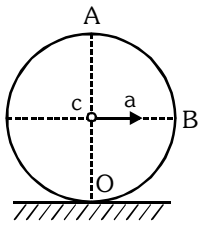
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Coll & Rot. Dy\Eng\04_RD\01_Eng.p65

Rolling kinematics

43. A point A is located on the rim of a wheel of radius R = 0.50 m which rolls without slipping along a horizontal surface with velocity v = 1.00 m/s. Find:
(a) the magnitude and the direction of the acceleration vector of the point A;
(b) the total distance s traversed by the point A between the two successive moments at which it touches the surface.

RD0410

44. A cylinder rolls without slipping over a horizontal plane. The radius of the cylinder is equal to r. Find the curvature radii of trajectories traced out by the points A and B (see Fig.).



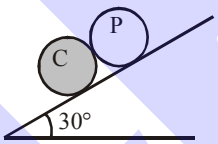
RD0411

Rolling dynamics

45. A solid uniform disk of mass m rolls down a fixed inclined plane without slipping with an acceleration a. Find the frictional force on the disk due to surface of the plane :

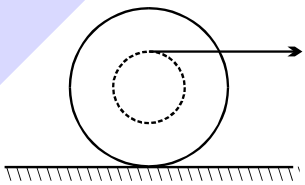
RD0019

46. A solid cylinder C and a hollow pipe P of same diameter are in contact when they are released from rest as shown in the figure on a long incline plane. Cylinder C and pipe P roll without slipping. Determine the clear gap (in m) between them after 6 seconds.



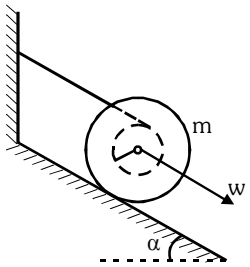
RD0020

47. A spool of inner radius R and outer radius 3R has a moment of inertia = MR^2 about an axis passing through its geometric centre, where M is the mass of the spool. A thread wound on the inner surface of the spool is pulled horizontally with a constant force = Mg. Find the acceleration of the point on the thread which is being pulled assuming that the spool rolls purely on the floor.



RD0021

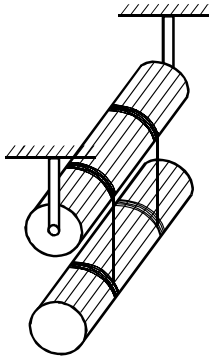
48. A spool with a thread wound on it is placed on an inclined smooth plane set at an angle $\alpha = 30°$ to the horizontal. The free end of the thread is attached to the wall as shown in Fig. The mass of the spool is m = 200 g, its moment of inertia relative to its own axis $\ell = 0.45$ g . m^2, the radius of the wound thread layer r = 3.0 cm. Find the acceleration of the spool axis.



RD0412

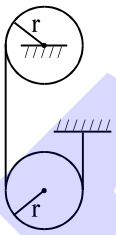
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, VPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

49. The arrangement shown in Fig. consists of two identical uniform solid cylinders, each of mass m, on which two light threads are wound symmetrically. Find the tension of each thread in the process of motion. The friction in the axle of the upper cylinder is assumed to be absent.



RD0413

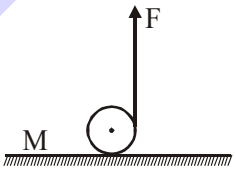
50. Two uniform cylinders, each of mass m = 10 kg and radius r = 150 mm, are connected by a rough belt as shown. If the system is released from rest, determine



(a) the velocity of the centre of cylinder A after it has moved through 1.2 m &
(b) the tension in the portion of the belt connecting the two cylinders.

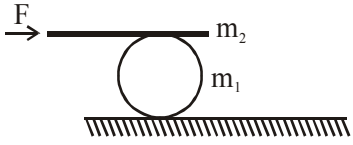
RD0037

51. The free end of the string wound on the surface of a solid cylinder of mass M & radius r is pulled up by a force F as shown. If there is sufficient friction between cylinder & floor so that the cylinder is able to roll without slipping, find the maximum angular acceleration that the cylinder can have (without cylinder leaving contact with surface).



RD0041

52. A man pushes a cylinder of mass m_1 with help of a plank of mass m_2 as shown. There is no slipping at any contact. the horizontal component of the force applied by the man is F. Find:



(a) the accelerations of the plank and the centre of mass of the cylinder, and
(b) the magnitudes and directions of frictional forces at contact points. **[JEE '99 , 6 + 4]**

RD0042

node06\B09C-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

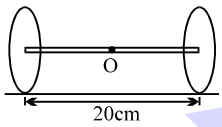
53. A 160 mm diameter pipe of mass 6 kg rests on a 1.5 kg plate. The pipe and plate are initially at rest when a force P of magnitude 25 N is applied for 0.75s. Knowing that $\mu_s = 0.25$ & $\mu_k = 0.20$ between the plate and both the pipe and the floor, determine;



(a) whether the pipe slides with respect to the plate .
(b) the resulting velocities of the pipe and of the plate .

RD0043

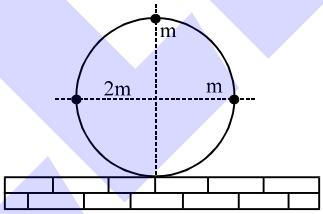
54. Two thin circular disks of mass 2kg and radius 10cm each are joined by a rigid massless rod of length 20cm. The axis of the rod is along the perpendicular to the planes of the disks through their centre. The object is kept on a truck in such a way that the axis of the object is horizontal and perpendicular to the direction of motion of the truck. Its friction with the floor of the truck is large enough so that the object can roll on the truck without slipping. Take x-axis as the direction of motion of the truck and z -axis as the vertically upwards direction. If the truck has an acceleration of $9 m/s^2$ calculate :



(a) the force of friction on each disk
(b) The magnitude and the direction of the frictional torque acting on each disk about the centre of mass O of the object. Express the torque in the vector form of unit vectors in the x-y and z directions.

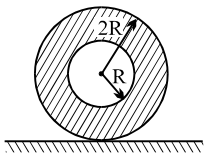
RD0044

55. A ring of mass m and radius R has three particles attached to the ring as shown in the figure. The centre of the ring has a speed v_0. Find the kinetic energy of the system. (Slipping is absent)



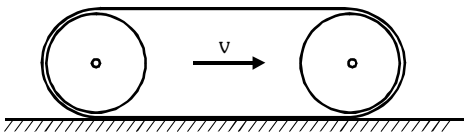
RD0022

56. A hollow cylinder with inner radius R, outer radius 2R and mass M is rolling with speed of its axis v. Its kinetic energy is :-



RD0023

57. Determine the kinetic energy of a tractor crawler belt of mass m if the tractor moves with velocity v.



RD0414

58. A uniform ball of radius r rolls without slipping down from the top of a sphere of radius R. Find the angular velocity of the ball at the moment it breaks off the sphere. The initial velocity of the ball is negligible.

RD0415

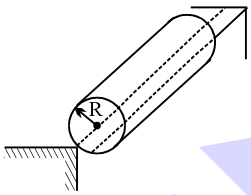
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Collision & Rot. Dy\Eng\04_RD\01_Eng.p65

59. Figure shows a rough track, a portion of which is in the form of a cylinder of radius R. With what minimum linear speed should a sphere of radius r be set rolling on the horizontal part so that it completely goes round the circle on the cylindrical part ?



RD0416

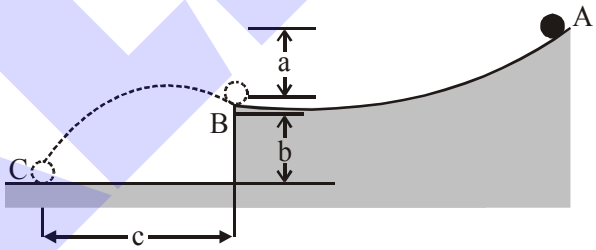
60. A rectangular rigid fixed block has a long horizontal edge. A solid homogeneous cylinder of radius R is placed horizontally at rest with its length parallel to the edge such that the axis of the cylinder and the edge of the block are in the same vertical plane as shown in figure. there is sufficient friction present at the edge so that a very small displacement cause the cylinder to roll of the edge without slipping. Determine



(a) the angle θ_C through which the cylinder rotates before it leaves contact with the edge,
(b) the speed of the centre of mass of the cylinder before leaving contact with the edge, and
(c) the ratio of the translational to rotational kinetic energies of the cylinder when its centre of mass is in horizontal line with the edge.

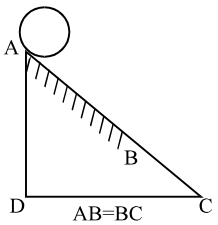
RD0045

61. A small sphere of mass m and radius r is released from rest at A and rolls without sliding on the curved surface to point B where it leaves the surface with a horizontal velocity. Knowing that a = 1.5 m & b = 1.2 m, determine;
(a) the speed of the sphere as it strikes the ground at C .
(b) the corresponding distance c .



RD0047

62. Portion AB of the wedge shown in figure is rough and BC is smooth. A solid cylinder rolls without slipping from A to B. Find the ratio of translational kinetic energy to rotational kinetic energy, when the cylinder reaches point C.

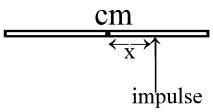


RD0052

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

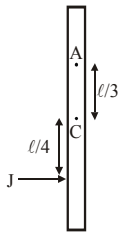
Angular momentum in rolling

63. A uniform rod of length l is given an impulse at right angle to its length as shown. Find the distance of instantaneous centre of rotation from the centre of the rod.



RD0024

64. A uniform rod of mass m and length ℓ is placed in gravity free space and linear impulse J is given to the rod at a distance x = ℓ/4 from centre 'C' and perpendicular to the rod. Point A is at a distance ℓ/3 from centre as shown in the figure. Then find



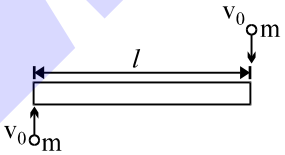
(i) Speed of centre of rod
(ii) Speed of point A
(iii) Speed of upper end of rod
(iv) Speed of lower end of rod

RD0025

65. A solid sphere of mass m and radius R is placed on a smooth horizontal surface. A sudden blow is given horizontally to the sphere at a height h = 4R/5 above the centre line. If I is the impulse of the blow then find
(a) the minimum time after which the highest point B will touch the ground
(b) the displacement of the centre of mass during this internal.

RD0026

66. On a smooth table two particles of mass m each, travelling with a velocity v_0 in opposite directions, strike the ends of a rigid massless rod of length l, kept perpendicular to their velocity. The particles stick to the rod after the collision.Find the tension in rod during subsequent motion.



RD0028

67. A uniform solid cylinder of mass m and radius R is set in rotation about its axis with an angular velocity ω_0, then lowered with its lateral surface onto a horizontal plane and released. The coefficient of friction between the cylinder and the plane is equal to k. Find:
(a) how long the cylinder will move with sliding;
(b) the total work performed by the sliding friction force acting on the cylinder.

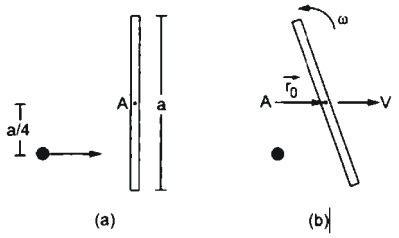
RD0417

68. A small disc and a thin uniform rod of length ℓ, whose mass is η times greater than the mass of the disc, lie on a smooth horizontal plane. The disc is set in motion, in horizontal direction and perpendicular to the rod, with velocity v, after which it elastically collides with the end of the rod. Find the velocity of the disc and the angular velocity of the rod after the collision. At what value of η will the velocity of the disc after the collision be equal to zero? Reverse its direction?

RD0418

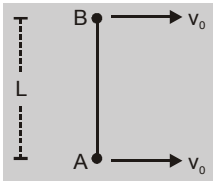
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, VPE, COM & Collision & Rot. Dy\Eng\04_RD\01_Eng.p65

69. A uniform rod of mass M and length a lies on a smooth horizontal plane. A particle of mass m moving at a speed v perpendicular to the length of the rod strikes it at a distance a/4 from the centre and stops after the collision. Find (a) the velocity of the centre of the rod and (b) the angular velocity of the rod about its centre just after the collision.



RD0419

70. Two small balls A and B, each of mass m, are joined rigidly to the ends of a light rod of length L (figure). The system translates on a frictionless horizontal surface with a velocity v_0 in a direction perpendicular to the rod. A particle P of mass m kept at rest on the surface sticks to the ball A as the ball collides with it. Find



(a) the linear speeds of the balls A and B after the collision.

(b) the velocity of the centre of mass C of the system A + B + P and

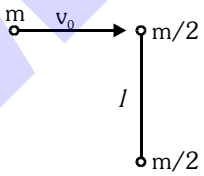
(c) the angular speed of the system about C after the collision

RD0420

71. A solid sphere rolling on a rough horizontal surface with a linear speed v collides elastically with a fixed, smooth, vertical wall. Find the speed of the sphere after it has started pure rolling in the backward direction.

RD0421

72. A ball of mass m moving with velocity v_0 experiences a head-on elastic collision with one of the spheres of a stationary rigid dumbbell as shown in Fig. The mass of each sphere equals m/2, and the distance between them is ℓ. Disregarding the size of the spheres, find the proper angular momentum $\tilde{M}$ of the dumbbell after the collision, i.e. the angular momentum in the reference frame moving translationally and fixed to the dumbbell's centre of inertia.



RD0422

73. A dumb-bell consists of two identical small balls of mass $\frac{1}{2}$ kg each connected to the two ends of a 50 cm. long light rod. The dumb-ball is rotating about a fixed axis through the centre of the rod and perpendicular to it at an angular speed of 10 rad/s. An impulsive force of average magnitude 5.0 N acts on one of the masses in the direction of its velocity for 0.10 s. Find the new angular velocity of the system.

RD0423

74. A uniform rod of mass m and length l is struck at an end by a force F perpendicular to the rod for a short time interval t. Calculate

(a) the speed of the centre of mass,

(b) the angular speed of the rod about the centre of mass,

(c) the kinetic energy of the rod and

(d) the angular momentum of the rod about the centre of the mass after the force has stopped to act. Assume that t is so small that the rod does not appreciably change its direction while the force acts.

RD0424

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

EXERCISE (JM)

1. A hoop of radius r and mass m rotating with an angular velocity ω_0 is placed on a rough horizontal surface. The initial velocity of the centre of the hoop is zero. What will be the velocity of the centre of the hoop when it ceases to slip? **[JEE Mains-2013]**

(1) $\frac{r\omega_0}{4}$ (2) $\frac{r\omega_0}{3}$ (3) $\frac{r\omega_0}{2}$ (4) $r\omega_0$

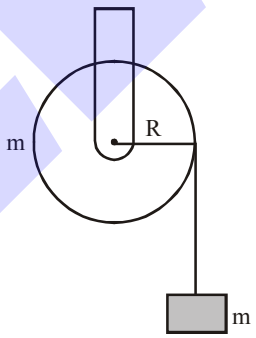
RD0158

2. A bob of mass m attached to an inextensible string of length ℓ is suspended from a vertical support. The bob rotates in a horizontal circle with an angular speed ω rad/s about the vertical. About the point of suspension : **[JEE Mains-2014]**

(1) Angular momentum changes in direction but not in magnitude
(2) Angular momentum changes both in direction and magnitude
(3) Angular momentum is conserved
(4) Angular momentum changes in magnitude but not in direction.

RD0159

3. A mass 'm' is supported by a massless string wound around a uniform hollow cylinder of mass m and radius R. If the string does not slip on the cylinder, with what acceleration will the mass fall on release? **[JEE Mains-2014]**



(1) $\frac{5g}{6}$ (2) g (3) $\frac{2g}{3}$ (4) $\frac{g}{2}$

RD0160

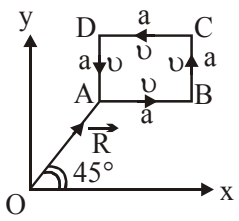
4. From a solid sphere of mass M and radius R a cube of maximum possible volume is cut. Moment of inertia of cube about an axis passing through its centre and perpendicular to one of its faces is:- **[JEE Mains-2015]**

(1) $\frac{4MR^2}{9\sqrt{3}\pi}$ (2) $\frac{4MR^2}{3\sqrt{3}\pi}$ (3) $\frac{MR^2}{32\sqrt{2}\pi}$ (4) $\frac{MR^2}{16\sqrt{2}\pi}$

RD0161

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, VPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

5. A particle of mass m is moving along the side of a square of side 'a', with a uniform speed υ in the x-y plane as shown in the figure : **[JEE Mains-2016]**



Which of the following statement is false for the angular momentum $\vec{L}$ about the origin ?

(1) $\vec{L}=\frac{m\upsilon}{\sqrt{2}}R\hat{k}$ when the particle is moving from D to A

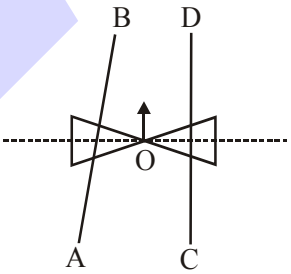
(2) $\vec{L}=-\frac{m\upsilon}{\sqrt{2}}R\hat{k}$ when the particle is moving from A to B

(3) $\vec{L}=m\upsilon\left[\frac{R}{\sqrt{2}}-a\right]\hat{k}$ when the particle is moving from C to D

(4) $\vec{L}=m\upsilon\left[\frac{R}{\sqrt{2}}+a\right]\hat{k}$ when the particle is moving from B to C

RD0162

6. A roller is made by joining together two cones at their vertices O. It is kept on two rails AB and CD which are placed asymmetrically (see figure), with its axis perpendicular to CD and its centre O at the centre of line joining AB and CD (see figure). It is given a light push so that it starts rolling with its centre O moving parallel to CD in the direction shown. As it moves, the roller will tend to :- **[JEE Mains-2016]**



(1) turn left and right alternately.
(2) turen left.
(3) turn right.
(4) go straight.

RD0163

7. The moment of inertia of a uniform cylinder of length ℓ and radius R about its perpendicular bisector is I. What is the ratio ℓ/R such that the moment of inertia is minimum ? **[JEE Main-2017]**

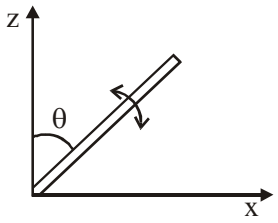
(1) 1 (2) $\frac{3}{\sqrt{2}}$ (3) $\sqrt{\frac{3}{2}}$ (4) $\frac{\sqrt{3}}{2}$

RD0164

node06\B09C-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

8. A slender uniform rod of mass M and length ℓ is pivoted at one end so that it can rotate in a vertical plane (see figure). There is negligible friction at the pivot. The free end is held vertically above the pivot and then released. The angular acceleration of the rod when it makes an angle θ with the vertical is :

[JEE Main-2017]

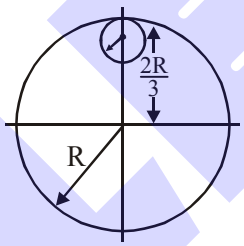


(1) $\frac{3g}{2\ell}\cos\theta$ (2) $\frac{2g}{3\ell}\cos\theta$ (3) $\frac{3g}{2\ell}\sin\theta$ (4) $\frac{2g}{3\ell}\sin\theta$

RD0165

9. From a uniform circular disc of radius R and mass 9M, a small disc of radius $\frac{R}{3}$ is removed as shown in the figure. The moment of inertia of the remaining disc about an axis perpendicular to the plane of the disc and passing through centre of disc is :

[JEE Main-2018]

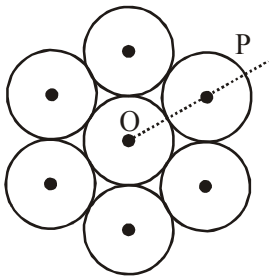


(1) $\frac{40}{9}MR^2$ (2) $10\ MR^2$ (3) $\frac{37}{9}MR^2$ (4) $4\ MR^2$

RD0166

10. Seven identical circular planar disks, each of mass M and radius R are welded symmetrically as shown. The moment of inertia of the arrangement about the axis normal to the plane and passing through the point P is :

[JEE Main-2018]



(1) $\frac{55}{2}\ MR^2$ (2) $\frac{73}{2}\ MR^2$ (3) $\frac{181}{2}\ MR^2$ (4) $\frac{19}{2}\ MR^2$

RD0167

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, VPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

SELECTED PROBLEMS FROM JEE-MAINS ONLINE PAPERS

11. To mop-clean a floor, a cleaning machine presses a circular mop of radius R vertically down with a total force F and rotates it with a constant angular speed about its axis. If the force F is distributed uniformly over the mop and if coefficient of friction between the mop and the floor is μ, the torque, applied by the machine on the mop is : **[JEE Main-2019_Jan]**

(1) $\frac{2}{3}\mu FR$ (2) μFR/3 (3) μFR/2 (4) μFR/6

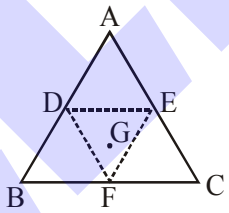
RD0242

12. The magnitude of torque on a particle of mass 1kg is 2.5 Nm about the origin. If the force acting on it is 1 N, and the distance of the particle from the origin is 5m, the angle between the force and the position vector is (in radians) :- **[JEE Main-2019_Jan]**

(1) $\frac{\pi}{8}$ (2) $\frac{\pi}{6}$ (3) $\frac{\pi}{4}$ (4) $\frac{\pi}{3}$

RD0243

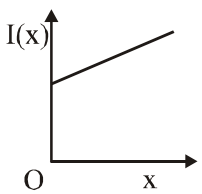
13. An equilateral triangle ABC is cut from a thin solid sheet of wood. (see figure) D, E and F are the mid-points of its sides as shown and G is the centre of the triangle. The moment of inertia of the triangle about an axis passing through G and perpendicular to the plane of the triangle is I_0. It the smaller triangle DEF is removed from ABC, the moment of inertia of the remaining figure about the same axis is I. Then: **[JEE Main-2019_Jan]**

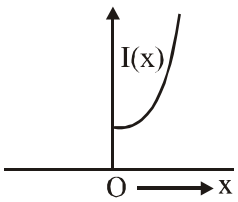


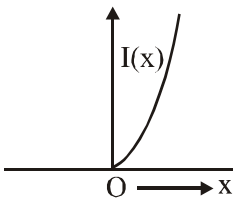
(1) $I=\frac{9}{16}I_0$ (2) $I=\frac{3}{4}I_0$ (3) $I=\frac{I_0}{4}$ (4) $I=\frac{15}{16}I_0$

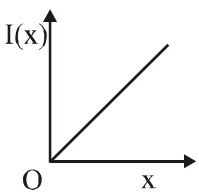
RD0244

14. The moment of inertia of a solid sphere, about an axis parallel to its diameter and at a distance of x from it, is I(x)'. Which one of the graphs represents the variation of I(x) with x correctlv? **[JEE Main-2019_Jan]**

(1) 

(2) 

(3) 

(4) 

RD0245

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

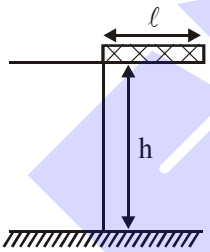
15. A solid sphere and solid cylinder of identical radii approach an incline with the same linear velocity (see figure). Both roll without slipping all throughout. The two climb maximum heights h_{sph} and h_{cyl} on the incline. The ratio $\frac{h_{sph}}{h_{cyl}}$ is given by :- **[JEE Main-2019_April]**



(1) $\frac{14}{15}$ (2) $\frac{4}{5}$ (3) 1 (4) $\frac{2}{\sqrt{5}}$

RD0246

16. A rectangular solid box of length 0.3 m is held horizontally, with one of its sides on the edge of a platform of height 5m. When released, it slips off the table in a very short time $\tau = 0.01$s, remaining essentially horizontal. The angle by which it would rotate when it hits the ground will be (in radians) close to :- **[JEE Main-2019_April]**



(1) 0.02 (2) 0.28 (3) 0.5 (4) 0.3

RD0247

17. A stationary horizontal disc is free to rotate about its axis. When a torque is applied on it, its kinetic energy as a function of θ, where θ is the angle by which it has rotated, is given as $k\theta^2$. If its moment of inertia is I then the angular acceleration of the disc is : **[JEE Main-2019_April]**

(1) $\frac{k}{2I}\theta$ (2) $\frac{k}{I}\theta$ (3) $\frac{k}{4I}\theta$ (4) $\frac{2k}{I}\theta$

RD0248

18. A solid sphere of mass M and radius R is divided into two unequal parts. The first part has a mass of $\frac{7M}{8}$ and is converted into a uniform disc of radius 2R. The second part is converted into a uniform solid sphere. Let I_1 be the moment of inertia of the disc about its axis and I_2 be the moment of inertia of the new sphere about its axis. The ratio I_1/I_2 is given by : **[JEE Main-2019_April]**

(1) 185 (2) 65 (3) 285 (4) 140

RD0249

19. A particle of mass m is moving along a trajectory given by

$x = x_0 + a\cos\omega_1 t$

$y = y_0 + b\sin\omega_2 t$

The torque, acting on the particle about the origin, at t = 0 is : **[JEE Main-2019_April]**

(1) $m(-x_0 b + y_0 a)\omega_1^2\hat{k}$ (2) $+my_0 a\omega_1^2\hat{k}$

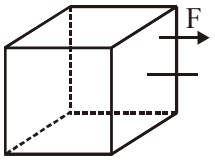
(3) $-m(x_0 b\omega_2^2 - y_0 a\omega_1^2)\hat{k}$ (4) Zero

RD0250

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, VPE, COM & Coll & Rot. Dy\Eng\04_RD\01_Eng.p65

20. Consider a uniform cubical box of side a on a rough floor that is to be moved by applying minimum possible force F at a point b above its centre of mass (see figure). If the coefficient of friction is μ = 0.4, the maximum possible value of $100 \times \frac{b}{a}$ for a box not to topple before moving is _____ .

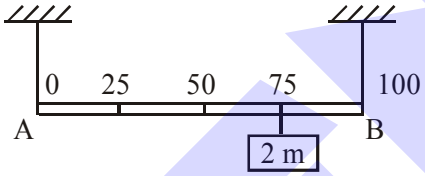
[JEE Main-2020_Jan]



RD0251

21. Shown in the figure is rigid and uniform one meter long rod AB held in horizontal position by two strings tied to its ends and attached to the ceiling. The rod is of mass 'm' and has another weight of mass 2 m hung at a distance of 75 cm from A. The tension in the string at A is :

[JEE Main-2020_Sep.]



(1) 2 mg (2) 0.5 mg (3) 0.75 mg (4) 1 mg

RD0252

22. A uniform cylinder of mass M and radius R is to be pulled over a step of height a (a < R) by applying a force F at its centre 'O' perpendicular to the plane through the axes of the cylinder on the edge of the step (see figure). The minimum value of F required is :

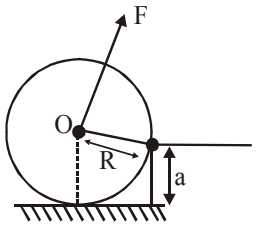
[JEE Main-2020_Sep.]

(1) $Mg\sqrt{1-\frac{a^2}{R^2}}$

(2) $Mg\sqrt{\left(\frac{R}{R-a}\right)^2-1}$

(3) $Mg\frac{a}{R}$

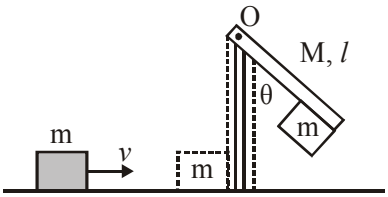
(4) $Mg\sqrt{1-\left(\frac{R-a}{R}\right)^2}$



RD0253

23. A block of mass m = 1 kg slides with velocity v = 6 m/s on a frictionless horizontal surface and collides with a uniform vertical rod and sticks to it as shown. The rod is pivoted about O and swings as a result of the collision making angle θ before momentarily coming to rest. If the rod has mass M = 2 kg, and length l = 1 m, the value of θ is approximately :
(Take g = 10 m/s^2)

[JEE Main-2020_Sep.]



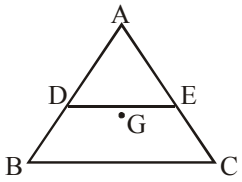
(1) 69° (2) 63° (3) 55° (4) 49°

RD0254

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

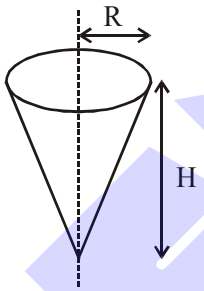
24. ABC is a plane lamina of the shape of an equilateral triagnle. D, E are mid points of AB, AC and G is the centroid of the lamina. Moment of inertia of the lamina about an axis passing through G and perpendicular to the plane ABC is I_0. If part ADE is removed, the moment of inertia of the remaining part about the same axis is $\frac{NI_0}{16}$ where N is an integer. Value of N is _____.

[JEE Main-2020_Sep.]



RD0255

25. Shown in the figure is a hollow icecream cone (it is open at the top). If its mass is M, radius of its top, R and height, H, then its moment of inertia about its axis is: **[JEE Main-2020_Sep.]**



(1) $\frac{MR^2}{2}$ (2) $\frac{MH^2}{3}$ (3) $\frac{MR^2}{3}$ (4) $\frac{M\left(R^2+H^2\right)}{4}$

RD0256

26. A uniform thin bar of mass 6 kg and length 2.4 meter is bent to make an equilateral hexagon. The moment of inertia about an axis passing through the centre of mass and perpendicular to the plane of hexagon is ____ $\times 10^{-1}$ kg m^2. **[JEE Main-2021_Feb.]**

RD0257

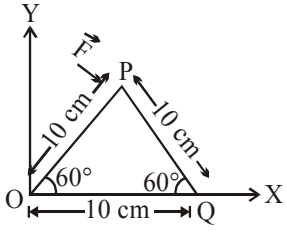
27. A triangular plate is shown. A force $\vec{F}=4\hat{i}-3\hat{j}$ is applied at point P. The torque at point P with respect to point 'O' and 'Q' are : **[JEE Main-2021_March]**

(1) $-15-20\sqrt{3}$, $15-20\sqrt{3}$

(2) $15+20\sqrt{3}$, $15-20\sqrt{3}$

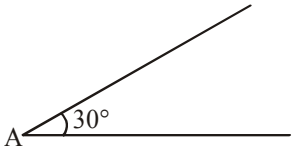
(3) $15-20\sqrt{3}$, $15+20\sqrt{3}$

(4) $-15+20\sqrt{3}$, $15+20\sqrt{3}$



RD0258

28. A sphere of mass 2kg and radius 0.5 m is rolling with an initial speed of 1 ms^{-1} goes up an inclined plane which makes an angle of 30° with the horizontal plane, without slipping. How low will the sphere take to return to the starting point A ? **[JEE Main-2021_March]**

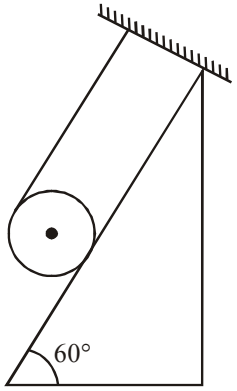


(1) 0.60 s (2) 0.52 s (3) 0.57 s (4) 0.80 s

RD0259

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Collision & Rot. Dy\Eng\04_RD\01_Eng.p65

29. A solid cylinder of mass m is wrapped with an inextensible light string and, is placed on a rough inclined plane as shown in the figure. The frictional force acting between the cylinder and the inclined plane is : **[JEE Main-2021_March]**



[The coefficient of static friction, μ_s, is 0.4]

(1) $\frac{7}{2}$mg (2) 5 mg (3) $\frac{mg}{5}$ (4) 0

RD0260

30. A rod of mass M and length L is lying on a horizontal frictionless surface. A particle of mass 'm' travelling along the surface hits at one end of the rod with a velocity 'u' in a direction perpendicular to the rod. The collision is completely elastic. After collision, particle comes to rest. The ratio of masses $\left(\frac{m}{M}\right)$ is $\frac{1}{x}$. The value of 'x' will be _____. **[JEE Main-2021_July]**

RD0261

31. Consider a situation in which a ring, a solid cylinder and a solid sphere roll down on the same inclined plane without slipping. Assume that they start rolling from rest and having identical diameter.
The correct statement for this situation is:- **[JEE Main-2021_July]**

(1) The sphere has the greatest and the ring has the least velocity of the centre of mass at the bottom of the inclined plane.
(2) The ring has the greatest and the cylinder has the least velocity of the centre of mass at the bottom of the inclined plane.
(3) All of them will have same velocity.
(4) The cylinder has the greatest and the sphere has the least velocity of the centre of mass at the bottom of the inclined plane.

RD0262

32. A particle of mass 'm' is moving in time 't' on a trajectory given by

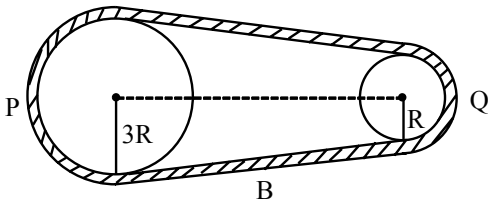
$$\vec{r} = 10\ \alpha t^2 \hat{i} + 5\beta(t-5)\hat{j}$$

Where α and β are dimensional constants.
The angular momentum of the particle becomes the same as it was for t = 0 at time t = _____seconds. **[JEE Main-2021_July]**

RD0263

node06\B09C-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

33. In the given figure, two wheels P and Q are connected by a belt B. The radius of P is three times as that of Q. In case of same rotational kinetic energy, the ratio of rotational inertias $\left(\frac{I_1}{I_2}\right)$ will be x : 1. The value of x will be _____. **[JEE Main-2021_July]**



RD0264

34. Two discs have moments of intertia I_1 and I_2 about their respective axes perpendicular to the plane and passing through the centre. They are rotating with angular speeds, ω_1 and ω_2 respectively and are brought into contact face to face with their axes of rotation coaxial. The loss in kinetic energy of the system in the process is given by : **[JEE Main-2021_Aug]**

(1) $\frac{I_1I_2}{(I_1+I_2)}(\omega_1-\omega_2)^2$ (2) $\frac{(I_1-I_2)^2\omega_1\omega_2}{2(I_1+I_2)}$ (3) $\frac{I_1I_2}{2(I_1+I_2)}(\omega_1-\omega_2)^2$ (4) $\frac{(\omega_1-\omega_2)^2}{2(I_1+I_2)}$

RD0265

35. Solid spherical ball is rolling on a frictionless horizontal plane surface about its axis of symmetry. The ratio of rotational kinetic energy of the ball to its total kinetic energy is :- **[JEE Main-2022_June]**

(1) $\frac{2}{5}$ (2) $\frac{2}{7}$ (3) $\frac{1}{5}$ (4) $\frac{7}{10}$

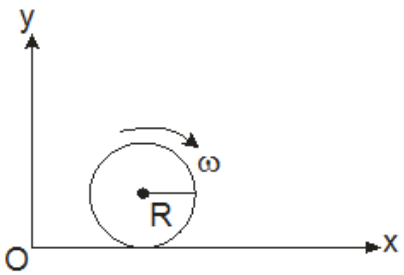
RD0425

36. A thin circular ring of mass M and radius R is rotating with a constant angular velocity 2 rads^{-1} in a horizontal plane about an axis vertical to its plane and passing through the center of the ring. If two objects each of mass m be attached gently to the opposite ends of a diameter of ring, the ring will then rotate with an angular velocity (in rads^{-1}). **[JEE Main-2022_June]**

(1) $\frac{M}{(M+m)}$ (2) $\frac{(M+2m)}{2M}$ (3) $\frac{2M}{(M+2m)}$ (4) $\frac{2(M+2m)}{M}$

RD0426

37. A spherical shell of 1 kg mass and radius R is rolling with angular speed ω on horizontal plane (as shown in figure). The magnitude of angular momentum of the shell about the origin O is $\frac{a}{3}R^2\omega$. The value of a will be : **[JEE Main-2022_June]**



(1) 2 (2) 3 (3) 5 (4) 4

RD0427

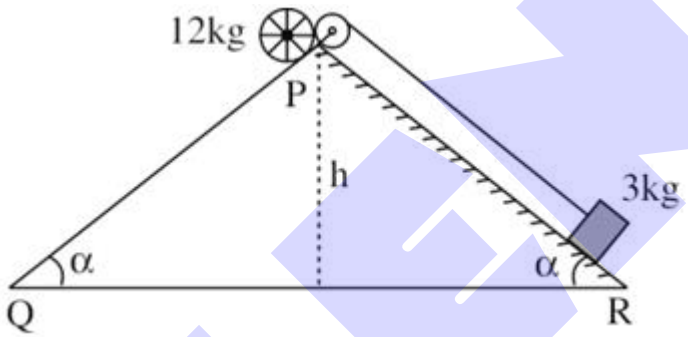
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, VPE, COM & Coll & Rot. Dy\Eng\04_RD\01_Eng.p65

38. The torque of a force $5\hat{i}+3\hat{j}-7\hat{k}$ about the origin is τ. If the force acts on a particle whose position vector is $2\hat{i}+2\hat{j}+\hat{k}$, then the value of τ will be : **[JEE Main-2022_July]**

(1) $11\hat{i}+19\hat{j}-4\hat{k}$ (2) $-11\hat{i}+9\hat{j}-16\hat{k}$ (3) $-17\hat{i}+19\hat{j}-4\hat{k}$ (4) $17\hat{i}+9\hat{j}+16\hat{k}$

RD0428

39. A rolling wheel of 12 kg is on an inclined plane at position P and connected to a mass of 3 kg through a string of fixed length and pulley as shown in figure. Consider PR as friction free surface. The velocity of centre of mass of the wheel when it reaches at the bottom Q of the inclined plane PQ will be $\frac{1}{2}\sqrt{xgh}$ m/s. The value of x is _____ . **[JEE Main-2022_June]**



RD0429

40. The moment of inertia of a uniform thin rod about a perpendicular axis passing through one end is I_1. The same rod is bent into a ring and its moment of inertia about a diameter is I_2. If $\frac{I_1}{I_2}$ is $\frac{x\pi^2}{3}$, then the value of x will be_____. **[JEE Main-2022_June]**

RD0430

41. A disc of mass 1 kg and radius R is free of rotate about a horizontal axis passing through its centre and perpendicular to the plane of disc. A body of same mass as that of disc is fixed at the highest point of the disc. Now the system is released, when the body comes to the lowest position, its angular speed will be $4\sqrt{\frac{x}{3R}}\text{rads}^{-1}$ where x = _____. (g = 10 ms^{-2}) **[JEE Main-2022_July]**

RD0431

42. A tube of length 50 cm is filled completely with an incompressible liquid of mass 250 g and closed at both ends. The tube is then rotated in horizontal plane about one of its ends with a uniform angular velocity $x\sqrt{F}$ rad s^{-1}. If F be the force exerted by the liquid at the other end then the value of x will be _____. **[JEE Main-2022_July]**

RD0432

node06\B09C-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

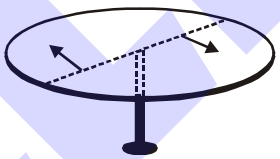
EXERCISE (JA)

1. A uniform circular disc of mass 50 kg and radius 0.4 m is rotating with an angular velocity of 10 rad s^{-1} about its own axis, which is vertical. Two uniform circular rings, each of mass 6.25 kg and radius 0.2 m, are gently placed symmetrically on the disc in such a manner that they are touching each other along the axis of the disc and are horizontal. Assume that the friction is large enough such that the rings are at rest relative to the disc and the system rotates about the original axis. The new angular velocity (in rad s^{-1}) of the system is **[IIT-JEE 2013]**

RD0180

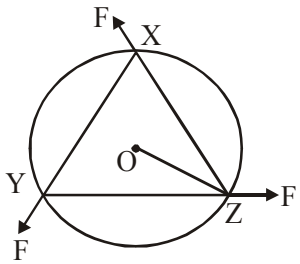
2. A horizontal circular platform of radius 0.5 m and mass 0.45 kg is free to rotate about its axis. Two massless spring toy-guns, each carrying a steel ball of mass 0.05 kg are attached to the platform at a distance 0.25 m from the centre on its either sides along its diameter (see figure). Each gun simultaneously fires the balls horizontally and perpendicular to the diameter in opposite directions. After leaving the platform, the balls have horizontal speed of 9 ms^{-1} with respect to the ground. The rotational speed of the platform in rad s^{-1} after the balls leave the platform is

[JEE Advanced-2014]



RD0181

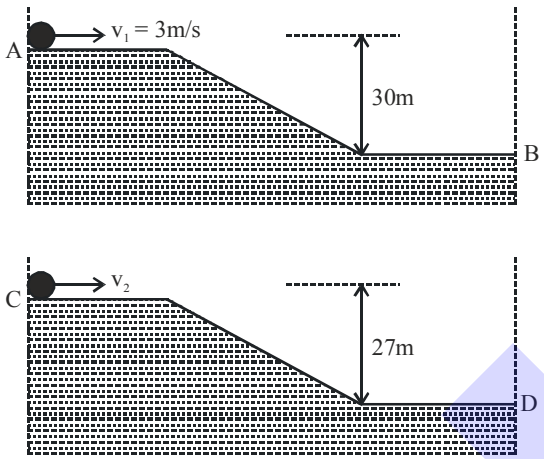
3. A uniform circular disc of mass 1.5 kg and radius 0.5 m is initially at rest on a horizontal frictionless surface. Three forces of equal magnitude F = 0.5 N are applied simultaneously along the three sides of an equilateral triangle XYZ with its vertices on the perimeter of the disc (see figure). One second after applying the forces, the angular speed of the disc in rad s^{-1} is: **[JEE Advanced-2014]**



RD0182

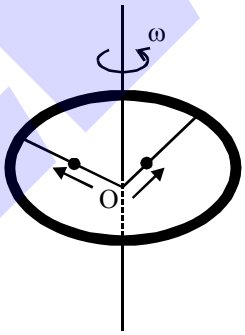
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, VPE, COM & Coll & Rot. Dy\Eng\04_RD\01_Eng.p65

4. Two identical uniform discs roll without slipping on two different, surfaces AB and CD (see figure) starting at A and C with linear speeds v_1 and v_2 respectively, and always remain in contact with the surfaces. If they reach B and D with the same linear speed and v_1 = 3m/s, then v_2 in m/s is ($g = 10\ m/s^2$) **[JEE Advanced-2015]**



RD0183

5. A ring of mass M and radius R is rotating with angular speed ω about a fixed vertical axis passing through its centre O with two point masses each of mass $\frac{M}{8}$ at rest at O. These masses can move radially outwards along two massless rods fixed on the ring as shown in the figure. At some instant the angular speed of the system is $\frac{8}{9}\omega$ and one of the masses is at a distance of $\frac{3}{5}$ R from O. At this instant the distance of the other mass from O is : **[JEE Advanced-2015]**



(A) $\frac{2}{3}R$ (B) $\frac{1}{3}R$ (C) $\frac{3}{5}R$ (D) $\frac{4}{5}R$

RD0184

6. The densities of two solid spheres A and B of the same radii R vary with radial distance r as $\rho_A(r) = k\left(\frac{r}{R}\right)$ and $\rho_B(r) = k\left(\frac{r}{R}\right)^5$, respectively, where k is a constant. The moments of inertia of the indivisual spheres about axes passing through their centres are I_A and I_B, respectively. If $\frac{I_B}{I_A} = \frac{n}{10}$, the value of n is. **[JEE Advanced-2015]**

RD0185

node06\B09C-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

7. A uniform wooden stick of mass 1.6 kg and length ℓ rests in an inclined manner on a smooth, vertical wall of height h ($<\ell$) such that a small portion of the stick extends beyond the wall. The reaction force of the wall on the stick is perpendicular to the stick. The stick makes an angle of 30° with the wall and the bottom of the stick is on a rough floor. The reaction of the wall on the stick is equal in magnitude to the reaction of the floor on the stick. The ratio h/ℓ and the frictional force f at the bottom of the stick are: ($g = 10\ ms^{-2}$) **[JEE Advanced-2016]**

(A) $\frac{h}{\ell}=\frac{\sqrt{3}}{16}, f=\frac{16\sqrt{3}}{3}N$

(B) $\frac{h}{\ell}=\frac{3}{16}, f=\frac{16\sqrt{3}}{3}N$

(C) $\frac{h}{\ell}=\frac{3\sqrt{3}}{16}, f=\frac{8\sqrt{3}}{3}N$

(D) $\frac{h}{\ell}=\frac{3\sqrt{3}}{16}, f=\frac{16\sqrt{3}}{3}N$

RD0186

8. The position vector $\vec{r}$ of a particle of mass m is given by the following equation $\vec{r}(t)=\alpha t^3\hat{i}+\beta t^2\hat{j}$, where $\alpha=\frac{10}{3}ms^{-3}$, $\beta = 5\ ms^{-2}$ and $m = 0.1$ kg. At $t = 1$ s, which of the following statement(s) is(are) true about the particle? **[JEE Advanced-2016]**

(A) The velocity $\vec{v}$ is given by $\vec{v}=(10\hat{i}+10\hat{j})ms^{-1}$

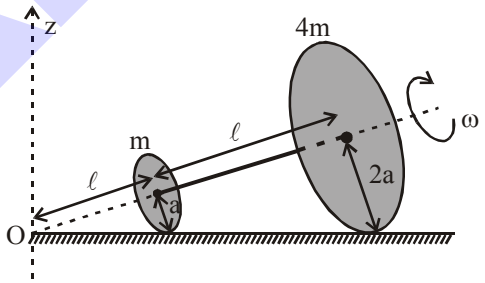
(B) The angular momentum $\vec{L}$ with respect to the origin is given by $\vec{L}=-\left(\frac{5}{3}\right)\hat{k}\ Nms$

(C) The force $\vec{F}$ is given by $\vec{F}=(\hat{i}+2\hat{j})N$

(D) The torque $\vec{\tau}$ with respect to the origin is given by $\vec{\tau}=-\left(\frac{20}{3}\right)\hat{k}\ Nm$

RD0187

9. Two thin circular discs of mass m and 4m, having radii of a and 2a, respectively, are rigidly fixed by a massless, right rod of length $\ell=\sqrt{24}a$ throught their center. This assembly is laid on a firm and flat surface, and set rolling without slipping on the surface so that the angular speed about the axis of the rod is ω. The angular momentum of the entire assembly about the point 'O' is $\vec{L}$ (see the figure). Which of the following statement(s) is(are) true ? **[JEE Advanced-2016]**



(A) The magnitude of angular momentum of the assembly about its center of mass is $17ma^2\omega/2$

(B) The magnitude of the z-component of $\vec{L}$ is $55\ ma^2\omega$

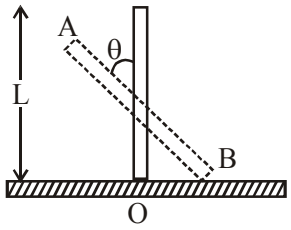
(C) The magnitude of angular momentum of center of mass of the assembly about the point O is $81ma^2\omega$

(D) The center of mass of the assembly rotates about the z-axis with an angular speed of ω/5

RD0188

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, VPE, COM & Collision & Rot. Dy\Eng\04_RD\01_Eng.p65

10. A rigid uniform bar AB of length L is slipping from its vertical position on a frictionless floor (as shown in the figure). At some instant of time, the angle made by the bar with the vertical is θ. Which of the following statements about its motion is/are correct ? **[JEE Advanced-2017]**



(A) When the bar makes an angle θ with the vertical, the displacement of its midpoint from the initial position is proportional to $(1 - \cos\theta)$

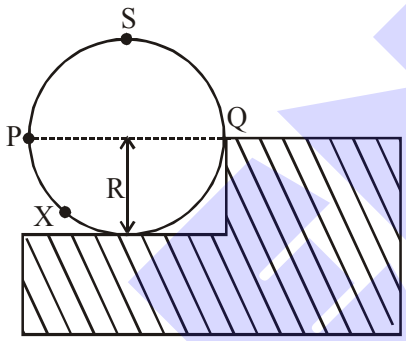
(B) The midpoint of the bar will fall vertically downward

(C) Instantaneous torque about the point in contact with the floor is proportional to $\sin\theta$

(D) The trajectory of the point A is a parabola

RD0190

11. A wheel of radius R and mass M is placed at the bottom of a fixed step of height R as shown in the figure. A constant force is continuously applied on the surface of the wheel so that it just climbs the step without slipping. Consider the torque τ about an axis normal to the plane of the paper passing through the point Q. Which of the following options is/are correct ? **[JEE Advanced-2017]**



(A) If the force is applied normal to the circumference at point X then τ is constant

(B) If the force is applied tangentially at point S then $\tau \neq 0$ but the wheel never climbs the step

(C) If the force is applied normal to the circumference at point P then τ is zero

(D) If the force is applied at point P tangentially then τ decreases continuously as the wheel climbs

RD0191

Paragraph for Question no. 12 & 13

One twirls a circular ring (of mass M and radius R) near the tip of one's finger as shown in Figure 1. In the process the finger never loses contact with the inner rim of the ring. The finger traces out the surface of a cone, shown by the dotted line. The radius of the path traced out by the point where the ring and the finger is in contact is r. The finger rotates with an angular velocity ω_0. The rotating ring rolls without slipping on the outside of a smaller circle described by the point where the ring and the finger is in contact (Figure 2). The coefficient of friction between the ring and the finger is μ and the acceleration due to gravity is g. **[JEE Advanced-2017]**

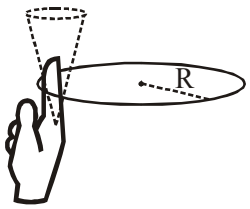


Figure 1

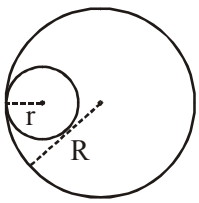


Figure 2

12. The total kinetic energy of the ring is :-

(A) $M\omega_0^2R^2$ (B) $M\omega_0^2(R-r)^2$ (C) $\frac{1}{2}M\omega_0^2(R-r)^2$ (D) $\frac{3}{2}M\omega_0^2(R-r)^2$

RD0192

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

13. The minimum value of ω_0 below which the ring will drop down is :-

(A) $\sqrt{\frac{3g}{2\mu(R-r)}}$ (B) $\sqrt{\frac{g}{\mu(R-r)}}$ (C) $\sqrt{\frac{2g}{\mu(R-r)}}$ (D) $\sqrt{\frac{2g}{2\mu(R-r)}}$

RD0192

14. The potential energy of a particle of mass m at a distance r from a fixed point O is given by $V(r) = kr^2/2$, where k is a positive constant of appropriate dimensions. This particle is moving in a circular orbit of radius R about the point O. If v is the speed of the particle and L is the magnitude of its angular momentum about O, which of the following statements is (are) true ?

[JEE Advanced-2018]

(A) $v = \sqrt{\frac{k}{2m}}R$ (B) $v = \sqrt{\frac{k}{m}}R$ (C) $L = \sqrt{mk}R^2$ (D) $L = \sqrt{\frac{mk}{2}}R^2$

RD0193

15. Consider a body of mass 1.0 kg at rest at the origin at time t = 0. A force $\vec{F} = (\alpha t\hat{i} + \beta\hat{j})$ is applied on the body, where $\alpha = 1.0\ Ns^{-1}$ and $\beta = 1.0\ N$. The torque acting on the body about the origin at time t = 1.0 s is $\vec{\tau}$. Which of the following statements is (are) true? **[JEE Advanced-2018]**

(A) $|\vec{\tau}| = \frac{1}{3}Nm$

(B) The torque $\vec{\tau}$ is in the direction of the unit vector $+\hat{k}$

(C) The velocity of the body at t = 1 s is $\vec{v} = \frac{1}{2}(\hat{i} + 2\hat{j})ms^{-1}$

(D) The magnitude of displacement of the body at t = 1 s is $\frac{1}{6}m$ **RD0194**

16. A ring and a disc are initially at rest, side by side, at the top of an inclined plane which makes an angle 60° with the horizontal. They start to roll without slipping at the same instant of time along the shortest path. If the time difference between their reaching the ground is $(2-\sqrt{3})/\sqrt{10}s$, then the height of the top of the inclined plane, in meters, is _____. Take $g = 10\ ms^{-2}$. **[JEE Advanced-2018]**

RD0195

17. A thin and uniform rod of mass M and length L is held vertical on a floor with large friction. The rod is released from rest so that it falls by rotating about its contact-point with the floor without slipping. Which of the following statement(s) is/are correct, when the rod makes an angle 60º with vertical ? [g is the acceleration due to gravity] **[JEE Advanced-2019]**

(1) The radial acceleration of the rod's center of mass will be $\frac{3g}{4}$

(2) The angular acceleration of the rod will be $\frac{2g}{L}$

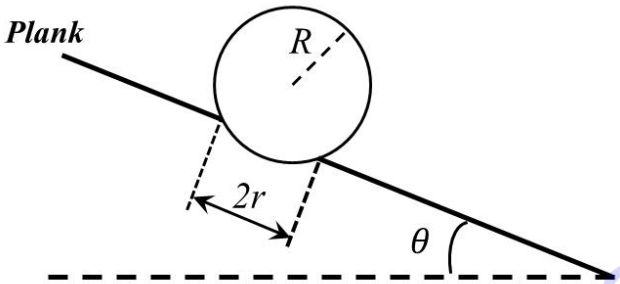
(3) The angular speed of the rod will be $\sqrt{\frac{3g}{2L}}$

(4) The normal reaction force from the floor on the rod will be $\frac{Mg}{16}$

RD0196

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, VPE, COM & Coll & Rot. Dy\Eng\04_RD\01_Eng.p65

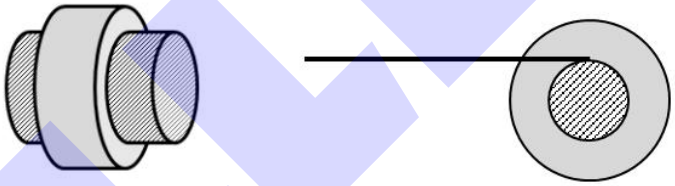
18. A football of radius R is kept on a hole of radius r (r < R) made on a plank kept horizontally. One end of the plank is now lifted so that it gets tilted making an angle θ from the horizontal as shown in the figure below. The maximum value of θ so that the football does not start rolling down the plank satisfies (figure is schematic and not drawn to scale) - **[JEE Advanced-2020]**



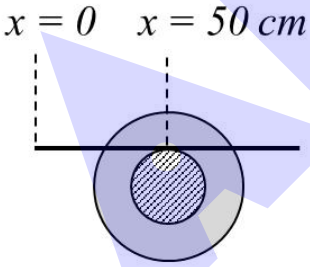
(A) $\sin\theta = \frac{r}{R}$ (B) $\tan\theta = \frac{r}{R}$ (C) $\sin\theta = \frac{r}{2R}$ (D) $\cos\theta = \frac{r}{2R}$

RD0266

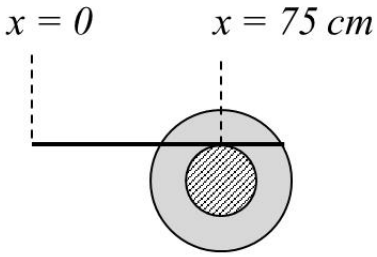
19. A small roller of diameter 20 cm has an axle of diameter 10 cm (see figure below on the left). It is on a horizontal floor and a meter scale is positioned horizontally on its axle with one edge of the scale on top of the axle (see figure on the right). The scale is now pushed slowly on the axle so that it moves without slipping on the axle, and the roller starts rolling without slipping. After the roller has moved 50 cm, the position of the scale will look like (figures are schematic and not drawn to scale)- **[JEE Advanced-2020]**



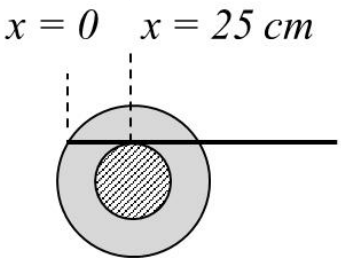
(A)



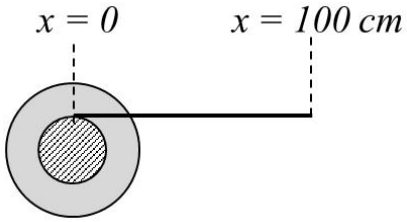
(B)



(C)



(D)



RD0267

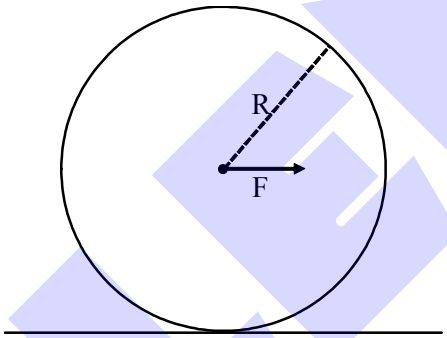
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Coll & Rot. Dy\Eng\04_RD\01_Eng.p65

20. Put a uniform meter scale horizontally on your extended index fingers with the left one at 0.00 cm and the right one at 90.00 cm. When you attempt to move both the fingers slowly towards the center, initially only the left finger slips with respect to the scale and the right finger does not. After some distance, the left finger stops and the right one starts slipping. Then the right finger stops at a distance x_R from the center (50.00 cm) of the scale and the left one starts slipping again. This happens because of the difference in the frictional forces on the two fingers. If the coefficients of static and dynamic friction between the fingers and the scale are 0.40 and 0.32, respectively, the value of x_R (in cm) is _____.- **[JEE Advanced-2020]**

RD0268

21. A horizontal force F is applied at the center of mass of a cylindrical object of mass m and radius R, perpendicular to its axis as shown in the figure. The coefficient of friction between the object and the ground is μ . The center of mass of the object has an acceleration a. The acceleration due to gravity is g. Given that the object rolls without slipping, which of the following statement(s) is(are) correct?

[JEE Advanced-2021]



(A) For the same F, the value of a does not depend on whether the cylinder is solid or hollow

(B) For a solid cylinder, the maximum possible value of a is 2 μg

(C) The magnitude of the frictional force on the object due to the ground is always μmg

(D) For a thin-walled hollow cylinder, $a = \frac{F}{2m}$ **RD0269**

22. A particle of mass M = 0.2 kg is initially at rest in the xy-plane at a point (x = –ℓ, y = –h), where ℓ = 10 m and h = 1m. The particle is accelerated at time t = 0 with a constant acceleration a = 10 m/s^2 along the positive x-direction. Its angular momentum and torque with respect to the origin, in SI units, are represented by $\vec{L}$ and $\vec{\tau}$, respectively. $\hat{i}, \hat{j}$ and $\hat{k}$ are unit vectors along the positive x, y and z-directions, respectively. If $\hat{k} = \hat{i} \times \hat{j}$ then which of the following statement(s) is(are) correct ?

[JEE Advanced-2021]

(A) The particle arrives at the point (x = ℓ, y = -h) at time t = 2s.

(B) $\vec{\tau} = 2\hat{k}$ when the particle passes through the point (x = ℓ, y = –h)

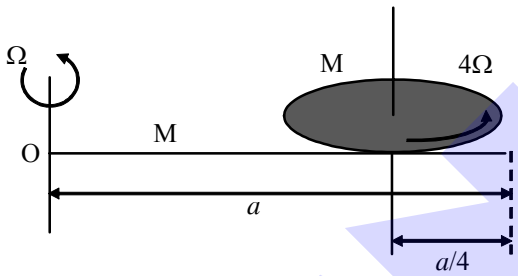
(C) $\vec{L} = 4\hat{k}$ when the particle passes through the point (x = ℓ, y = –h)

(D) $\vec{\tau} = \hat{k}$ when the particle passes through the point (x = 0, y = –h)

RD0270

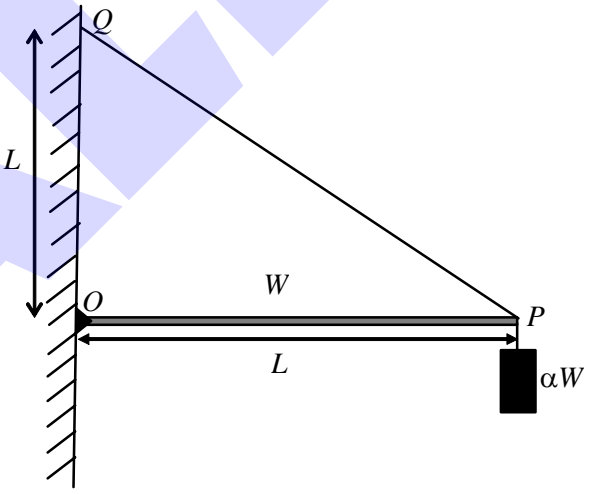
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, VPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

23. A thin rod of mass M and length a is free to rotate in horizontal plane about a fixed vertical axis passing through point O. A thin circular disc of mass M and of radius a/4 is pivoted on this rod with its center at a distance a/4 from the free end so that it can rotate freely about its vertical axis, as shown in the figure. Assume that both the rod and the disc have uniform density and they remain horizontal during the motion. An outside stationary observer finds the rod rotating with an angular velocity Ω and the disc rotating about its vertical axis with angular velocity 4Ω. The total angular momentum of the system about the point O is $\left(\frac{Ma^2\Omega}{48}\right)n$. The value of n is___. **[JEE Advanced-2021]**



RD0271

24. One end of a horizontal uniform beam of weight W and length L is hinged on a vertical wall at point O and its other end is supported by a light inextensible rope. The other end of the rope is fixed at point Q, at a height L above the hinge at point O. A block of weight αW is attached at the point P of the beam, as shown in the figure (not to scale). The rope can sustain a maximum tension of $(2\sqrt{2})$W. Which of the following statement(s) is(are) correct ? **[JEE Advanced-2021]**



(A) The vertical component of reaction force at O does not depend on α
(B) The horizontal component of reaction force at O is equal to W for α = 0.5
(C) The tension in the rope is 2W for α = 0.5
(D) The rope breaks if α > 1.5

RD0272

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

Question Stem for Question Nos. 25 and 26

A pendulum consists of a bob of mass m = 0.1 kg and a massless inextensible string of length L = 1.0 m. It is suspended from a fixed point at height H = 0.9 m above a frictionless horizontal floor. Initially, the bob of the pendulum is lying on the floor at rest vertically below the point of suspension. A horizontal impulse P = 0.2 kg-m/s is imparted to the bob at some instant. After the bob slides for some distance, the string becomes taut and the bob lifts off the floor. The magnitude of the angular momentum of the pendulum about the point of suspension just before the bob lifts off is J kg-m^2/s. The kinetic energy of the pendulum just after the lift-off is K Joules.

[JEE Advanced-2021]

25. The value of J is ____.

RD0433

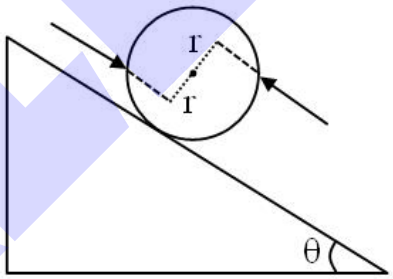
26. The value of K is ____.

RD0434

27. At time t = 0, a disk of radius 1 m starts to roll without slipping on a horizontal plane with an angular acceleration of $a = \frac{2}{3} rads^{-2}$. A small stone is stuck to the disk. At t = 0, it is at the contact point of the disk and the plane. Later, at time $t = \sqrt{\pi}\ s$ the stone detaches itself and flies off tangentially from the disk. The maximum height (in m) reached by the stone measured from the plane is $\frac{1}{2} + \frac{x}{10}$. The value of x is_____. [Take g = 10 m s^{-2}.] **[JEE Advanced-2022]**

RD0435

28. A solid sphere of mass 1 kg and radius 1 m rolls without slipping on a fixed inclined plane with an angle of inclination θ = 30° from the horizontal. Two forces of magnitude 1N each, parallel to the incline, act on the sphere, both at distance r = 0.5 m from the center of the sphere, as shown in the figure. The acceleration of the sphere down the plane is_____ms^{-2}. (Take g = 10 m s^{-2}.)



[JEE Advanced-2022]

RD0436

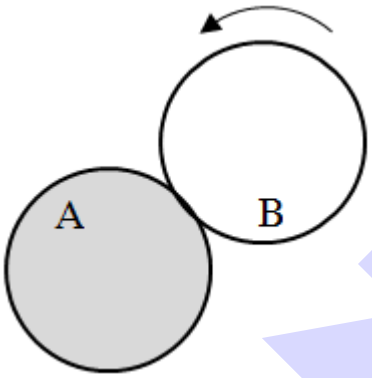
29. A particle of mass 1 kg is subjected to a force which depends on the position as $\vec{F} = -k\left(x\hat{i} + y\hat{j}\right) kgms^{-2}$ with k = 1 kgs^{-2}. At time t = 0, the particle's position $\vec{r} = \left(\frac{1}{\sqrt{2}}\hat{i} + \sqrt{2}\hat{j}\right) m$ and its velocity $\vec{v} = \left(-\sqrt{2}\hat{i} + \sqrt{2}\hat{j} + \frac{2}{\pi}\hat{k}\right) ms^{-1}$. Let v_x and v_y denote the x and the y components of the particle's velocity, respectively. **Ignore gravity.** When z = 0.5 m, the value of $(xv_y - yv_x)$ is_____ m^2s^{-1}. **[JEE Advanced-2022]**

RD0437

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, VPE, COM & Coll & Rot. Dy\Eng\04_RD\01_Eng.p65

30. A flat surface of a thin uniform disk A of radius R is glued to a horizontal table. Another thin uniform disk B of mass M and with the same radius R rolls without slipping on the circumference of A, as shown in the figure. A flat surface of B also lies on the plane of the table. The center of mass of B has fixed angular speed about the vertical axis passing through the center of A. The angular momentum of B is $nM\omega R^2$ with respect to the center of A. Which of the following is the value of n?

[JEE Advanced-2022]



(A) 2 (B) 5 (C) $\frac{7}{2}$ (D) $\frac{9}{2}$

RD0438

ALLEN®

node06\B09C-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

ANSWER KEY

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	B	C	B	A	C	B	D	D	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	B	A	C	A	A	A	D	C	A
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	A	A	B	A	B	D	A	D	A	B
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	C	C	A	A	A	C	C	A	B	D
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	C	A	D	A	D	B	B	C	D	C
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	C	A	C	B	A	B	C	D	C	B
Que.	61	62	63	64	65	66	67	68	69	70
Ans.	D	A	C	D	D	B	C	D	A	B
Que.	71	72	73	74	75	76	77	78	79	80
Ans.	A	D	B	B	D	C	D	A	B	B
Que.	81	82	83	84	85	86	87	88	89	90
Ans.	A	B	A	D	A	B	B	D	A	B
Que.	91	92	93	94	95	96	97	98	99	100
Ans.	A	B	A	A	B	A	D	A	B	D
Que.	101	102	103	104	105	106	107	108	109	110
Ans.	A	A	D	B	B	B	A	B	A	B
Que.	111	112	113	114	115	116	117	118	119	120
Ans.	A	A	C	B	C	A	C	A	A	D
Que.	121	122	123	124	125	126	127	128	129	130
Ans.	D	A	B	A	B	B	A	D	B	D
Que.	131	132	133	134	135	136	137	138	139	140
Ans.	A	B	D	B	A	C	B	A	C	A
Que.	141	142	143	144	145	146	147	148	149	150
Ans.	D	C	A	A	A	B	A	A	D	B
Que.	151	152	153	154	155	156	157	158	159	160
Ans.	A	A,B,C,D	B,C	A,D	B,D	A,C	A,D	A,C,D	A,C,D	B,C
Que.	161	162	163	164	165	166	167	168	169	170
Ans.	A,B,C,D	A,C,D	A,B	A,B,C	A,C	A	A	D	B	A-R; B-P; C-T; D-S
Que.	171									
Ans.	A-S,T; B-R; C-P,R,T; D-P,Q									

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, VPE, COM & Coll & Rot. Dy\Eng\04_RD\01_Eng.p65

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	C	C	C	A	C	C	B,C,D	C	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	A,B,D	D	B	B	C	C	C	B	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	A,B,C	B,D	C	A-R; B-S; C-P; D-Q	B,C,D	D	B	C	A	A,C
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	A,C	B,C,D	A,B,C	C	A,B,C	C	D	B	C	B
Que.	41	42	43	44	45					
Ans.	B,C,D	A,C,D	B	C	D					

EXERCISE # S

1. 2 **2.** $\frac{ml^2}{12}$ **3.** $57/140MR^2$ **4.** $\frac{a\ell^4}{36}$ **5.** $2\pi\left(\frac{Aa^4}{4}+\frac{Ba^5}{5}\right)$

6. 8 kg **7.** 30° **8.** **98 N, 65 N** **9.** **(a)** 740 N-m; **(b)** 590 N vertical and 120 N horizontal

10. $(3/8)\,F_g$ **11.** (a) 5 m/s² →, (b) $0.3 < h < 1.5$ m **12.** 10^{-3} rad/s² clockwise **13.** **2**

14. $\frac{L}{3}$ **15.** (a) $\frac{9g}{7}\downarrow$ (b) $\frac{4mg}{7}\uparrow$ **16.** (i) $10/13 m/s^2$, (ii) $5000/26\pi$,(iii) 480/13N **17.** 2N

18. $T=\frac{1}{2}mg$, $w_0 = gmr^2/I$ **19.** $\alpha = 2mgx/R\ell\,(M+2m)$ **20.** **100 rad/s.**

21. **8.4 rad/s²** **22.** **0.89 m/s²** **23.** 1.63 N, 1.224 m **24.** $M=2m\left(\frac{2gh}{R^2\omega^2}-1\right)$

25. $\sqrt{5gR}$ **26.** (a) $\omega=\sqrt{\frac{6g}{\ell}}$, (b) $T=\frac{11mg}{4}$ **27.** $\omega=\sqrt{\frac{6g}{\ell}}$, $\sqrt{6g\ell}$ **28.** $\sqrt{15}$ ft/sec

29. $\frac{\omega}{3}$ **30.** $\frac{4\pi}{5}$ **31.** $T=\frac{mr_0^2v_0^2}{r^3}$, $\frac{1}{2}mv_0^2\left[\frac{r_0^2}{r^2}-1\right]$ **32.** $m = 2kr_1^2/v_2^2$

33. **(a)** $\omega=\frac{I_1\omega_1+I_2\omega_2}{I_1+I_2}$; **(b)** $A=-\frac{I_1I_2}{2(I_1+I_2)}(\omega_1-\omega_2)^2$ **34.** **41°** **35.** $t = 2/\sqrt{5}$ sec, $\theta = 4\pi/5$ rad

36. (a) $\omega/3$, (b) $\frac{\sqrt{37}}{3}m\omega R$, (c) $\frac{\sqrt{37}}{3}m\omega R$ **37.** 75 J **38.** $\frac{mv}{4}$

39. $\omega'_1=\frac{I_1\omega_1r_2+I_2\omega_2r_1}{I_2r_1^2+I_1r_2^2}r_2$; $\omega'_2=\frac{I_1\omega_1r_2+I_2\omega_2r_1}{I_2r_1^2+I_1r_2^2}r_1$ **40.** **(a)** 1.0 N-m **(b)** 9.0 kJ **(c)** 60 kg-m²/s

41. (a) $\ell = 0.1$m; (b) $\omega' = 1$rad/s ; (c) laminar sheet will never come to rest

42. $M=\frac{1}{2}mgv_0t^2\cos\alpha$; $M = (mv_0^3/2g)\sin2\alpha\cos\alpha = 36.27$ kg.m²/s

43. **(a)** $w_A = v^2/R = 2.0$ m/s², the vector w_A is permanently directed to the centre of the wheel; **(b)** $s = 8R = 4.0$m.

44. **$R_A = 4r$, $R_B = 2\sqrt{2}\,r$** **45.** **1/2 ma** **46.** **15** **47.** **16 m/s²**

48. $w = g\sin\alpha/(1 + I/mr^2) = 1.42$ m/s² **49.** **$T=\frac{1}{10}mg$** **50.** (a) $4\sqrt{\frac{3}{7}}$ m/s, (b) $\frac{200}{7}$ N **51.** $\alpha_{max}=\frac{2g}{3r}$

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, WPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65

52. **(a)** $a_c = \frac{4F}{3m_1 + 8m_2}$, $a_p = 2a_c$ **; (b)** friction at the top of the cylinder = $3m_1F(3m_1+8m_2)$ towards right; friction at the bottom = $m_1F/(3m_1+8m_2)$ towards right.

53. **(a)** pipe rolls without sliding **(b)** pipe : 5/6 m/s →, 125/12 rad/s (anticlockwise); plate : 5/3 m/s →

54. $6N, -0.6\hat{j} \pm 0.6\hat{k}$ **55.** $6mv_0^2$ **56.** $\frac{13}{16}Mv^2$ **57.** $T = mv^2$ **58.** $\omega = \sqrt{10g(R+r)/17r^2}$

59. $\sqrt{\frac{27}{7}g(R-r)}$ **60.** (a) $\theta_C = \cos^{-1}(4/7)$, (b) $v = \sqrt{4/7\ gR}$, (c) $K_T/K_R = 6$

61. **(a)** 6.68 m/s **(b)** 2.27 m **62.** **5** **63.** $\frac{l^2}{12x}$ **64.** (i) $\frac{J}{m}$ (ii) zero (iii) $\frac{J}{2m}$ (iv) $\frac{5}{2}\frac{J}{m}$

65. (a) $t = \frac{\pi Rm}{2I}$; (b) $s = \frac{\pi R}{2}$ **66.** $\frac{2mv_0^2}{l}$ **67.** (a) $t = \frac{1}{3}\omega_0 R/kg$; (b) $A = -\frac{1}{6}m\omega_0^2R^2$

68. $v' = v(4-\eta)/(4+\eta)$, $\omega = 12v/\ell(4+\eta)$. For $\eta = 4$ and $\eta > 4$ **69.** **(a)** $\frac{m}{M}v$; (b) $\omega = \frac{3mv}{Ma}$

70. **(a)** $\frac{v_0}{2}, v_0$ **(b)** $\frac{2}{3}v_0$ **along the initial motion of the rod (c)** $\frac{v_0}{2L}$ **71. 3v/7**

72. $\tilde{M} = \frac{1}{3}\ell mv_0$ **73. 12 rad/s** **74. (a)** $\frac{Ft}{m}$ **(b)** $\frac{6Ft}{m\ell}$ **(c)** $\frac{2F^2t^2}{m}$ **(d)** $\frac{F\ell t}{2}$

EXERCISE # JM

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	1	4	1	1,3	2	3	3	4	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	2	4	2	1	3	4	4	2	75
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	4	4	2	11	1	8	1	3	3	4
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	1	10	9	3	2	3	3	3	3	8
Que.	41	42								
Ans.	5	4								

EXERCISE # JA

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	8	4	2	7	C,D	6	D	A,B,D	D	A,B,C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B,C or C	BONUS	B	B,C	A,C	0.75 (0.74,0.76)	1,3,4	A	B	25.6
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B,D	A,B,C	49	A,B,D	0.18	0.16	0.48-0.56	2.80-2.92	3	B

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\2-Circular Motion, VPE, COM & Colli & Rot. Dy\Eng\04_RD\01_Eng.p65